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A scalable multidimensional fully implicit solver for Hall magnetohydrodynamics

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Abstract

We propose an optimally performant fully implicit algorithm for the Hall magnetohydrodynamics (HMHD) equations based on multigrid-preconditioned Jacobian-free Newton-Krylov methods. HMHD is a challenging system to solve numerically because it supports stiff fast dispersive waves. The preconditioner is formulated using an operator-split approximate block factorization (Schur complement), informed by physics insight. We use a vector-potential formulation (instead of a magnetic field one) to allow a clean segregation of the problematic $\nabla\times\nabla\times$ operator in the electron Ohm's law subsystem. This segregation allows the formulation of an effective damped block-Jacobi smoother for multigrid. We demonstrate by analysis that our proposed block-Jacobi iteration is convergent and has the smoothing property. The resulting HMHD solver is verified linearly with wave propagation examples, and nonlinearly with the GEM challenge reconnection problem by comparison against another HMHD code. We demonstrate the excellent algorithmic and parallel performance of the algorithm up to 16384 MPI tasks in two dimensions.

Keywords: implicit methods, Hall magnetohydrodynamics, Newton-Krylov, multigrid, physics-based preconditioning, approximate block factorization.

1. Introduction

We propose a scalable implicit algorithm for the Hall magnetohydrodynamics (HMHD) model, which is of relevance in the modeling of astrophysical, magnetospheric, and laboratory (e.g., fusion) plasmas. HMHD extends the standard MHD model by allowing independent motion between ions and electrons within so-called ion inertial layers [1] thereby fundamentally altering their topological properties, enhancing magnetic-field transport within them and impacting important basic plasma physics phenomena such as magnetic reconnection [2]. HMHD also allows for electron demagnetization within electron inertial layers, which is necessary to model vacuum regions with MHD properly [1, 3].

HMHD is well known to support fast dispersive waves, with frequencies growing quadratically with the number of spatial mesh points, resulting in explicit Courant-Friedrichs-Lewy (CFL) timestep stability constraints scaling quadratically with the mesh spacing (rather than linearly for non-dispersive waves). As a result, the explicit time integration of HMHD becomes very expensive for sufficiently refined meshes, encouraging the exploration of implicit timestepping schemes. After spatial discretization, however, implicit schemes require finding the solution of a large system of nonlinear algebraic equations, which are particularly challenging to solve iteratively in an efficient manner (e.g., with multilevel methods such as multigrid) for stiff hyperbolic PDE systems such as HMHD. The key roadblock for multigrid deployment is the formulation of effective multigrid smoothers.

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Several authors have attempted to build implicitness in the solution algorithm of the HMHD equations over the years. The earliest study we are aware of is Ref. [4], where a semi-implicit solver for the HMHD system based on a pseudospectral discretization in cylindrical geometry was proposed for out-of-plane uniform magnetic fields (which incidentally do not support the problematic stiff hyperbolic waves mentioned earlier). The stability analysis of the semi-implicit time-advance formulation in the general case (including stiff hyperbolic waves) was studied in Ref. [5], concluding that strict time-centering was needed for numerical stability. Several authors have explored fully implicit HMHD formulations, either in its reduced form (i.e., assuming plasma incompressibility) [6, 7, 8, 9, 10] or in primitive-variable form [8, 11, 12, 13, 14, 15]. Of these, many of them employ either direct solvers [7, 8, 9, 14, 15] or incomplete LU factorizations [13], which are known not to scale well either algorithmically under grid refinement, or in parallel (and in fact little information is provided about either scaling in the references). Scalable fully implicit solutions for the reduced HMHD system [6, 10] and the primitive-variable HMHD system in a non-standard ion formulation [11] were proposed that show significant promise. However, the latter formulation (relevant to the present study) has been found by the author to be more prone to numerical noise from fast dispersive waves than the more standard electron formulation proposed in this study, and problematic to solve when attempting to include high-order dissipation (via hyperresistivity, mimicking electron viscosity [2]) to control it.[1] Other authors have attempted to bypass the numerical stiffness challenges of HMHD either by adding a fictitious displacement current term in Ampere's law [16, 17] or by reverting back to a multi-fluid formulation with promising approximate block-factorization preconditioners [18, 19]. The former approach is problematic because the formulation is physically inconsistent, and can be shown to introduce instability [17]. The latter introduces a large number of additional equations, and features fast charge-separation and lightwave timescales that complicate an implicit treatment (although these could be important in some physical contexts such as the treatment of boundary sheaths).

This paper builds on earlier work [20] where a scalable, parallel, fully implicit, fully nonlinear solver for the 3D resistive compressible MHD equations was demonstrated, and augments it to include electron Hall and pressure effects. As in the reference, we base our nonlinear solver approach on the Newton-Raphson iterative algorithm. Krylov iterative techniques [21], implemented Jacobian-free [22, 23] (i.e., without ever forming and storing the Jacobian matrix) for memory efficiency, are employed for the required algebraic matrix inversions. Here, FGMRES (Flexible Generalized Minimal RESiduals [24, 25]) is employed as the Krylov solver of choice, due to the lack of symmetry in the algebraic system of interest. The flexible character of FGMRES allows the preconditioner to change between successive GMRES iterations. We will exploit this feature in our implementation.

The efficiency and scalability of Krylov methods depends strongly on adequate preconditioning [21]. In Refs. [6, 20], a suitable multigrid-based preconditioner is developed for the MHD system around the parabolization concept, whereby a hyperbolic system is reformulated as a parabolic one using a Schur factorization, which is in turn amenable to a multilevel treatment. The parabolization procedure is generalized here to HMHD. Using physics insight, we approximate the exact Schur factorization with an operator-split procedure, where the stiff electron subsystem (which we will term electron MHD, or EMHD) is inverted first, followed by the ion MHD subsystem inversion proposed in Ref. [20]. All subsystems are solved using a fully parallel geometric multigrid (MG) implementation based on a coarse-grid parallel-domain agglomeration strategy (similar to that proposed in Refs. [26, 27]). For the EMHD subsystem (which is the stiffest one in HMHD), we propose a novel six-equation block damped-Jacobi smoothing strategy that is shown by

[1] As discussed in [2], the fast-dispersive-wave noise issue in HMHD in the absence of hyperresistivity originates in the fact that HMHD dispersive wave frequencies scale as $\omega \sim k^2$, with k the wavenumber, but so does the visco-resistive damping rate, γ. Therefore no dissipation transition lengthscale exists for which the hyperbolic regime (with $\omega > \gamma$) transitions into a dissipative one (with $\omega < \gamma$). Defining a transition lengthscale necessitates higher-order dissipation, as discussed later in this study.

analysis to lead to convergent Jacobi iterations and to possess the smoothing property. Damped Jacobi is demonstrated in Ref. [12] to be a viable smoother for a semi-implicit discretization of the 2D EMHD system (without hyperresistivity), and is implemented in a nonlinear MG solver. However, no grid-convergence study is offered in the reference, and relatively small implicit timesteps (of $\mathcal{O}(40)$ explicit CFLs) are used. Here, we extend the analysis to general multidimensional geometry (which requires dealing with the $\nabla\times\nabla\times$ operator, very challenging for geometric MG solvers), include hyperresistivity, and push implicit timesteps to 8400 CFLs without loss of algorithmic performance. The formulation is enabled by a vector-potential formulation of the HMHD system (instead of the more common magnetic-field one). The resulting smoother is able to deal seamlessly with both the whistler-wave and the fourth-order hyper-resistivity operators, both featuring $\nabla\times\nabla\times$ terms. As a result of these advances, the preconditioner renders the Jacobian-free Newton-Krylov (JFNK) solver optimally scalable algorithmically and in parallel (in a weak scaling sense) up to the largest number of processors considered in this study (16384 processors).

The rest of the paper is organized as follows. Section 2 introduces the base model equations. Sec. 3 discusses the details of the proposed physics-based preconditioner. Numerical details of our implementation, including the multigrid (MG) treatment of the hyperbolically stiff HMHD system, are provided in Sec. 4 (with detailed analysis of the smoothing properties of our block-Jacobi treatment of the Hall MHD equations provided in Appendix C). Linear and nonlinear verification results of our implementation using wave propagation examples and the GEM challenge problem [2], respectively, along with a weak parallel scalability study of our implicit solver up to 16384 cores, are presented in Sec. 5. Finally, we conclude in Sec. 6.

2. Vector-potential formulation of the Hall MHD model

In Alfvénic units (i.e, Alfvén speed $v_A = B_0/\sqrt{\mu_0\rho_0}$, Alfvén time $\tau_A = L/v_A$, where B_0, ρ_0, and L are the reference magnetic field, density, and length, respectively), the HMHD model of interest for this study reads:

$$\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\mathbf{v}) = 0, \tag{1}$$

$$\frac{\partial\mathbf{B}}{\partial t} + \nabla\times\mathbf{E} = \mathbf{0} \tag{2}$$

$$\frac{\partial(\rho\mathbf{v})}{\partial t} + \nabla\cdot\Big[\rho\mathbf{v}\mathbf{v} - \mathbf{B}\mathbf{B} + \mathbb{I}(p+\frac{B^2}{2}) + \Pi\Big] = \mathbf{0}, \tag{3}$$

$$\frac{\partial p}{\partial t} + \nabla\cdot(\mathbf{v}^* p) + (\gamma-1)p\nabla\cdot\mathbf{v}^* = (\gamma-1)(Q-\nabla\cdot\mathbf{q}), \tag{4}$$

with $\mathbb{I}$ the identity dyad, ρ the particle density, $\mathbf{v}$ the center-of-mass plasma velocity (which is approximated here as the ion velocity, i.e., $\mathbf{v}\approx\mathbf{v}_i$, which will have implications later on), $\mathbf{B}$ the magnetic field, $\mathbf{E}$ the electric field, $p = \rho T = (1+\alpha_T)\rho T_e$ the total pressure, $\Pi = \sum_{s=i,e}\Pi_s$ the total viscous stress tensor, and $\mathbf{v}^* = \frac{\alpha_T\mathbf{v}+\mathbf{v}_e}{\alpha_T+1} = \mathbf{v} - \frac{d_i}{1+\alpha_T}\frac{\mathbf{j}}{\rho}$. Here, $\alpha_T = T_i/T_e$ is a fixed ion-to-electron temperature ratio, $\mathbf{v}_e = \mathbf{v} - d_i\mathbf{j}/\rho$ is the electron velocity, $\mathbf{j} = \nabla\times\mathbf{B}$ is the current, and $d_i = c/\omega_{pi}L$ the dimensionless ion inertial length. The velocity $\mathbf{v}^*$ in the pressure equation, Eq. 4, is derived by adding the ion and electron internal energy evolution equations [28], and using the fixed-temperature-ratio assumption to ensure existence of a local energy conservation law for fixed temperature ratio α, as we outline below. Note that the pressure equation reverts to the electron pressure equation $p_e = p/(1+\alpha_T)$ for cold ions ($\alpha_T \ll 1$), and to the ion pressure equation $p_i = p\alpha_T/(1+\alpha_T)$ for cold electrons ($\alpha_T \gg 1$).

In these equations, η is the normalized resistivity (inverse Lundquist number), $\mathbf{q}$ is the heat flux, and $Q = \eta j^2 - \sum_{s=i,e}\Pi_s : \nabla\mathbf{v}_s$ features the joule and viscous heating sources. Simple closures for the heat

flux $\mathbf{q} = -\kappa \nabla T_e$ and the viscous stress tensors $\Pi_s = -\rho \nu_s \nabla \mathbf{v}_s$ (with s the species, and ν_s the normalized viscosity) are considered at this time. We neglect the electron viscous stress term in the equation of motion (Eq. 3), since it is physically much smaller in magnitude than the ion one [28]. Anisotropic heat transport has been implemented in our code-base [29], but will not be considered here. More accurate closures for viscous stresses (such as gyroviscosity) will be considered in future work.

Key to the character of the model in Eqs. 1-4 is the specification of the electric field $\mathbf{E}$ via an Ohm's law. In general, the Ohm's law is derived from the electron equation of motion, which reads:

$$\mathbf{E} = -\mathbf{v}\times\mathbf{B} + \eta\mathbf{j} + \frac{d_i}{\rho}(\mathbf{j}\times\mathbf{B} - \nabla p_e - \nabla\cdot\Pi_e) + d_e^2\frac{d}{dt}\left(\frac{\mathbf{j}}{\rho}\right) = -\mathbf{v}_e\times\mathbf{B} + \eta\mathbf{j} - \frac{d_i}{\rho}(\nabla p_e + \nabla\cdot\Pi_e) + d_e^2\frac{d}{dt}\left(\frac{\mathbf{j}}{\rho}\right). \tag{5}$$

Here, the last term takes into account the effects of electron inertia, and can be expanded as:

$$\frac{d}{dt}\left(\frac{\mathbf{j}}{\rho}\right) = \partial_t\left(\frac{\mathbf{j}}{\rho}\right) + \mathbf{v}_e\cdot\nabla\left(\frac{\mathbf{j}}{\rho}\right).$$

This term replaces the more physically intuitive one, $d\mathbf{v}_e/dt$ to enable a rigorous energy conservation principle with the $\mathbf{v}_i \approx \mathbf{v}$ assumption. The parameter d_i characterizes the importance of Hall effects: for $d_i \ll 1$ the model transitions to resistive MHD, while for $d_i \gg 1$ the model transitions to the so-called electron MHD model [30, 31]. The parameter $d_e = c/\omega_{pe}L$ is the dimensionless electron inertial length, which characterizes the importance of electron inertia and regularizes HMHD in vacuum regions ($d_e \gg 1$). The ion and electron inertial lengths are related as $d_e^2/d_i^2 = m_e/m_i$.

The system of Eqs. 1-4, together with Eq. 5, admits the following local energy conservation law (Appendix A):

$$\frac{\partial}{\partial t}\left(\rho\frac{v^2}{2} + \frac{B^2}{2} + \frac{p}{\gamma-1} + d_e^2\frac{j^2}{2\rho}\right) + \nabla\cdot\left[\rho\mathbf{v}\frac{v^2}{2} + \mathbf{E}\times\mathbf{B} + \mathbf{q} + \sum_{s=i,e}\mathbf{v}_s\cdot\Pi_s + \frac{\gamma}{\gamma-1}p\mathbf{v}^* + d_e^2\mathbf{v}_e\frac{j^2}{2\rho}\right] = 0. \tag{6}$$

This implies that, with appropriate boundary conditions (e.g., perfect conductor ($\mathbf{E}\times\mathbf{n} = 0$), impenetrable wall ($\mathbf{v}\cdot\mathbf{n} = 0$), no viscous stress ($\mathbf{n}\cdot\Pi_s = \mathbf{0}$), no normal heat flux ($\mathbf{n}\cdot\mathbf{q} = 0$), etc.), the total energy is exactly conserved. Numerically, energy will be conserved only approximately, since we are solving for the specific internal energy (Eq. 4) instead of total energy (the former is more advantageous for low plasma-beta applications, with the plasma beta defined as $\beta_p = 2\mu_0 p_0/B_0^2$), and we are neglecting the electron viscous stress tensor contribution in Eq. 3. In our simulations, energy is typically conserved to better than one part in 10^3.

For sufficiently large d_i, the system in Eqs. 1-4, together with Eq. 5, is dispersively hyperbolic, supporting fast dispersive normal modes (Chap. 3 in Ref. [32]). In many applications of interest, however, dynamical time scales of interest are much slower than those associated with these normal modes. This is so because macroscopic dynamical time scales of interest are typically related to the bulk plasma, and thus controlled by ions, while the much faster normal modes are related to electron physics. In such instances, an implicit approach that steps over the fast normal-mode time scales to resolve the slower time scales of interest is advantageous. This is the subject of this paper. However, as we shall see, the extreme hyperbolic character of the Hall MHD model makes the task of developing an optimal, scalable solver a very difficult task.

For preconditioning purposes, we will find it beneficial to consider a governing equation for the magnetic vector potential $\mathbf{A}$ instead of the magnetic field $\mathbf{B}$ (which are related as $\mathbf{B} = \nabla\times\mathbf{A}$). Evolving $\mathbf{A}$ instead of $\mathbf{B}$ allows more flexibility in the numerical discretization without sacrificing the solenoidal property of the magnetic field. However, it requires a suitable gauge, and care must be taken with boundary conditions. The

evolution equation for $\mathbf{A}$ is found from $\mathbf{E} = -\nabla\Phi - \partial_t \mathbf{A}$ (where Φ is a scalar potential) and the generalized Ohm's law, Eq. 5, and reads:

$$\frac{\partial \mathbf{A}}{\partial t} + \mathbf{E} = \frac{\partial \mathbf{A}}{\partial t} - \mathbf{v}_e \times \mathbf{B} + \eta \mathbf{j} - \frac{d_i}{\rho}(\nabla p_e + \nabla \cdot \Pi_e) + d_e^2 \frac{d}{dt}\left(\frac{\mathbf{j}}{\rho}\right) = -\nabla\Phi. \tag{7}$$

One can eliminate Φ by choosing an appropriate gauge. Here, we consider the Weyl gauge, $\Phi = 0$, to find (Appendix B):

$$\frac{\partial \mathbf{A}}{\partial t} - \mathbf{v}_e \times (\nabla \times \mathbf{A}) + \eta \nabla \times \nabla \times \mathbf{A} - \frac{d_i}{\rho}(\nabla p_e + \nabla \cdot \Pi_e) + \frac{d_e^2}{\rho}(\partial_t \mathbf{j} + \nabla \cdot (\mathbf{v}_e \mathbf{j})) = 0. \tag{8}$$

Because we are solving for the vector potential, the solenoidal constraint of the magnetic field is automatically guaranteed as long as $\nabla \cdot \nabla \times = 0$ discretely (as is the case in our implementation [33]).

As stated earlier, a simple viscous closure is used for both species, $\Pi_s \approx -\rho \nu_s \nabla \mathbf{v}_s$. It is important to note that the main purpose of keeping the electron viscous tensor contribution in our implementation, in the absence of a more informed closure, is to regularize the electron equation of motion by providing some dissipation scale. Such regularization is needed for numerical purposes, as has been described in various other studies [2, 6]. In our context, we have seen rapid degradation of the preconditioner performance when such a term is lacking. The reason is that approximations introduced in the preconditioner impact small scales the most [20], and the dispersive nature of Hall MHD seems to amplify these differences to the point that it renders the preconditioner ineffective after just a few implicit time steps.

For electrons, the simple viscous closure gives:

$$\nabla \cdot \Pi_e = -\nabla \cdot (\rho \nu_e \nabla \mathbf{v}_e) = -\nabla \cdot \left[\rho \nu_e \nabla \left(\mathbf{v} - \frac{d_i}{\rho}\mathbf{j}\right)\right] = -\nabla \cdot \left[\rho \nu_e \nabla \left(\mathbf{v} - \frac{d_i}{\rho}\nabla \times \nabla \times \mathbf{A}\right)\right], \tag{9}$$

where ν_e is the electron viscosity (also known as hyperresistivity). This expression contains a fourth-order dissipation operator on $\mathbf{A}$, with $d_i \nabla \cdot \Pi_e \sim d_i^2 \nu_e \nabla^2 (\nabla \times \nabla \times \mathbf{A})$. The higher differential order is needed to provide a dissipation length scale to quadratic dispersive waves with dispersion relation $\omega \sim k^2$ [2]. Numerical considerations can help to determine a grid-bound ν_e that provides sufficient dissipation while avoiding excessive stiffness from the high-order operators. An estimate of ν_e can be obtained by balancing the damping rate $d_i^2 \nu_e k^4$ with the dispersive wave frequency $d_i v_A k_\parallel k$, to find [6]:

$$\nu_e = C d_i^{-1} v_A k_\parallel k^{-3}, \tag{10}$$

with C a constant of order unity. We discuss the specifics of the numerical implementation of the electron viscous tensor term in the next section.

We comment briefly on the discretization of Eqs. 1-4. Spatially, the system is discretized using second-order finite volumes with all variables (vectors and scalars) co-located at cell centers [33]. Such spatial representation is conservative and nonlinearly stable, and has proved to remarkably robust both in resistive [33] and Hall MHD [8]. Temporally, we employ either a θ-scheme with $\theta = 0.5$ (second-order Crank-Nicolson), or second-order backward differentiation formulas (BDF2) [34], depending on the problem. Either approach results in a set of nonlinear algebraic equations $\mathbf{G}(\mathbf{x}) = \mathbf{0}$, with $\mathbf{x}^T = (\rho, T, \mathbf{A}, \mathbf{v})$, that needs to be inverted every time step. For this, we employ preconditioned Jacobian-free Newton-Krylov (JFNK) methods. The next section discusses our approach to preconditioning the Hall MHD system.

3. Physics-based preconditioning strategy

We formulate our preconditioner by first parabolizing the implicitly discretized hyperbolic system via an approximate block factorization, and then invert the resulting systems with multigrid methods (MG) for scalability.

As preconditioners, MG methods have been shown in many applications to lead to optimal convergence rates in JFNK [35]. The key element in a working MG solver is the smoother. While smoothers can be found fairly easily for diagonally dominant systems (in a point or block sense; p. 96 in [36]), it is remarkably hard otherwise. Hyperbolic systems (such as MHD) can be shown to be diagonally submissive when time steps larger than the explicit CFL stability constraint are employed (see Ref. [37] for an in-depth explanation of this issue). However, hyperbolic systems can be conveniently parabolized in an implicit time-stepping setting. The basic idea is to produce a diagonally dominant parabolic operator from an ill-conditioned hyperbolic system of equations by a block factorization [38, 35, 39, 37, 6, 40, 20, 41, 42, 43, 44]. The procedure can be understood easily with a first-order coupled hyperbolic linear system:

$$\begin{aligned} \partial_t u &= \partial_x v, \\ \partial_t v &= \partial_x u. \end{aligned}$$

Differencing implicitly in time (with backward Euler for simplicity) but keeping the continuum spatial representation, we have:

$$u^{n+1} = u^n + \Delta t \partial_x v^{n+1}, \tag{11}$$
$$v^{n+1} = v^n + \Delta t \partial_x u^{n+1}. \tag{12}$$

It is now possible to substitute the second equation into the first to obtain the following parabolic equation:

$$(\mathbb{I} - \Delta t^2 \partial_{xx}) u^{n+1} = u^n + \Delta t \partial_x v^n, \tag{13}$$

which is equivalent to Eqs. 11-12, but much better conditioned for MG because the parabolic operator is diagonally dominant. Once u^{n+1} is found, v^{n+1} can be found straightforwardly from Eq. 12. A similar idea is behind the method of differential approximations [45], and several semi-implicit solvers developed in the MHD context [46, 47, 4].

The parabolization of the hyperbolic equation is a manifestation of the implicit discretization of the semi-discrete hyperbolic system. Indeed, hyperbolic waves feature a finite propagation speed. By construction, implicit methods are designed to step over stiff hyperbolic waves to follow the dynamics on a slower time scale. In order to do so, the implicit scheme must change the character of the equations for unresolved hyperbolic time scales, and parabolization is the outcome. Resolved hyperbolic time scales maintain their character. This can be clearly seen from Eq. 13: for small enough time steps, the second-order term can be neglected, and one recovers an explicit (hyperbolic) formulation. It is only for large enough time steps that the system changes character to become parabolic.

The connection between parabolization and the Schur factorization [6, 20] enables the generalization of these ideas to more complicated hyperbolic systems. Equations 11-12, when written in block-matrix form, can be factorized as:

$$\begin{bmatrix} \mathbb{I} & -\Delta t \partial_x \\ -\Delta t \partial_x & \mathbb{I} \end{bmatrix} = \begin{bmatrix} \mathbb{I} & -\Delta t \partial_x \\ 0 & \mathbb{I} \end{bmatrix} \begin{bmatrix} \mathbb{I} - \Delta t^2 \partial_x^2 & 0 \\ 0 & \mathbb{I} \end{bmatrix} \begin{bmatrix} \mathbb{I} & 0 \\ -\Delta t \partial_x & \mathbb{I} \end{bmatrix}.$$

The connection is now obvious, as the parabolic operator appears naturally. Its usefulness will become apparent in the following sections.

3.1. Block structure of the linearized Hall MHD model

We begin by considering the linearized Hall MHD model for our preconditioning development. Neglecting heat sources (which do not introduce stiff time scales), the coupling structure of the linearized

MHD system in terms of the linear updates $\delta\rho$, δT_e, $\delta\mathbf{A}$, and $\delta\mathbf{v}$ reads:

$$\begin{aligned}
\delta\rho &= \mathscr{L}_\rho(\delta\rho,\delta\mathbf{v}),\\
\delta p &= \mathscr{L}_T(\delta p,\delta\mathbf{v},\delta\mathbf{A},\delta\rho),\\
\delta\mathbf{A} &= \mathscr{L}_E(\delta\mathbf{v},\delta\mathbf{A},\delta\rho,\delta p),\\
\delta\mathbf{v} &= \mathscr{L}_v(\delta\mathbf{v},\delta\mathbf{A},\delta\rho,\delta p),
\end{aligned}$$

where the $\mathscr{L}_i$ represent linear operators. The resulting Jacobian structure is:

$$J\delta\mathbf{x} = \begin{bmatrix} D_\rho & 0 & 0 & U_{\rho\mathbf{v}} \\ \boxed{L_{p\rho}} & D_p & \boxed{U_{p\mathbf{A}}} & U_{p\mathbf{v}} \\ \boxed{L_{\mathbf{A}\rho}} & \boxed{L_{\mathbf{A}p}} & \boxed{D_\mathbf{A}} & U_{\mathbf{vA}} \\ L_{\mathbf{v}\rho} & L_{\mathbf{v}p} & L_{\mathbf{vA}} & D_\mathbf{v} \end{bmatrix} \begin{pmatrix} \delta\rho \\ \delta p \\ \delta\mathbf{A} \\ \delta\mathbf{v} \end{pmatrix}. \tag{14}$$

The boxed blocks contain Hall-MHD contributions from finite d_i terms (stemming from the Ohm's law and the definition of $\mathbf{v}^*$ in the temperature equation). Here, diagonal blocks are given by:

$$\begin{aligned}
D_\rho\delta\rho &= \frac{\delta\rho}{\Delta t} + \theta[\nabla\cdot(\mathbf{v}_0\delta\rho) - D\nabla^2\delta\rho],\\
D_p\delta p &= \left(\frac{1}{\Delta t} + \theta(\gamma-1)\nabla\cdot\mathbf{v}_0^*\right)\delta p + \theta[\nabla\cdot(\mathbf{v}_0^*\delta p) - \frac{\kappa}{\alpha_T\rho_0}\nabla^2\delta T],\\
D_\mathbf{A}\delta\mathbf{A} &= \frac{\delta\mathbf{A}^*}{\Delta t} - \theta\left(\mathbf{v}_{e,0}\times\nabla\times\delta\mathbf{A}^* - d_i\frac{\nabla\times\nabla\times\delta\mathbf{A}}{\rho_0}\times\mathbf{B_0} - \eta\nabla\times\nabla\times\delta\mathbf{A} + \nabla\cdot\left[\rho_0 d_i\nu_e\nabla\left(\frac{\nabla\times\nabla\times\delta\mathbf{A}}{\rho_0}\right)\right]\right),\\
D_\mathbf{v}\delta\mathbf{v} &= \rho_0\left[\frac{\delta\mathbf{v}}{\Delta t} + \theta(\mathbf{v}_0\cdot\nabla\delta\mathbf{v} + \delta\mathbf{v}\cdot\nabla\mathbf{v}_0)\right] - \theta\nabla\cdot(\nu_i\rho_0\nabla\delta\mathbf{v}),
\end{aligned}$$

where, for $D_\mathbf{v}$, we have used the non-conservative form of the momentum equation [20]. Here, θ is the time centering parameter, and $\delta\mathbf{A}^* = \delta\mathbf{A} + \frac{d_e^2}{\rho_0}\nabla\times\nabla\times\delta\mathbf{A}$ (obtained by linearizing Eq. B.4 in Appendix B). Off-diagonal blocks L and U are given by (using the non-conservative form of the momentum equation, and ignoring additional Gauge-related $\mathscr{O}(d_e^2)$ terms in the vector-potential equation):

$$\begin{aligned}
U_{\rho\mathbf{v}}\delta\mathbf{v} &= \theta\nabla\cdot(\rho_0\delta\mathbf{v}),\\
U_{p\mathbf{A}}\delta\mathbf{A} &= -\theta\frac{d_i}{1+\alpha_T}\left[\nabla\cdot(\frac{p_0}{\rho_0}\nabla\times\nabla\times\delta\mathbf{A}) + (\gamma-1)p_0\nabla\cdot\frac{\nabla\times\nabla\times\delta\mathbf{A}}{\rho_0}\right],\\
U_{\mathbf{vA}}\delta\mathbf{v} &= -\theta\delta\mathbf{v}\times\mathbf{B}_0,\\
U_{p\mathbf{v}}\delta\mathbf{v} &= \theta[\nabla\cdot(\delta\mathbf{v}p_0) + (\gamma-1)p_0\nabla\cdot\delta\mathbf{v}],\\
L_{\mathbf{A}\rho}\delta\rho &\approx \theta\frac{d_i}{1+\alpha_T}\frac{\delta\rho}{\rho_0^2}\nabla p_0,\\
L_{p\rho}\delta\rho &= \theta\frac{d_i}{1+\alpha_T}\left[\nabla\cdot\left(\frac{\delta\rho}{\rho_0^2}p_0\mathbf{j}_0\right) + (\gamma-1)p_0\nabla\cdot\left(\frac{\delta\rho}{\rho_0^2}\mathbf{j}_0\right)\right],\\
L_{\mathbf{A}p}\delta p &= -\theta\frac{d_i}{1+\alpha_T}\frac{1}{\rho_0}\nabla\delta p,\\
L_{\mathbf{v}\rho}\delta\rho &= \delta\rho\left(\frac{\mathbf{v}_0}{\Delta t} + \theta\mathbf{v}_0\cdot\nabla\mathbf{v}_0\right) - \theta\nabla\cdot(\nu_i\delta\rho\nabla\mathbf{v}_0),\\
L_{\mathbf{vA}}\delta\mathbf{A} &= -\theta\left[\mathbf{j}_0\times\nabla\times\delta\mathbf{A} + (\nabla\times\nabla\times\delta\mathbf{A})\times\mathbf{B}_0\right],\\
L_{\mathbf{v}p}\delta p &= \theta\nabla\delta p.
\end{aligned}$$

3.2. Parabolization of the Hall MHD Jacobian matrix

Upon inspection of Eq. 14, it is clear that the the convenient "arrow structure" of the resistive MHD Jacobian matrix [20] has been spoiled. However, the stiff electron timescales introduced in Hall MHD are all contained in the magnetic field diagonal block $D_{\mathbf{A}}$ via the $d_i(\frac{\nabla\times\nabla\times\delta\mathbf{A}}{\rho_0}\times\mathbf{B_0})$ term, responsible for the propagation of fast whistler waves. This realization opens the way for a tractable physics-based preconditioning algorithm.

The Jacobian recovers a manageable structure by simply neglecting the block $U_{p\mathbf{A}}$ (stemming from the linearization of the current correction in the $\mathbf{v}^*$ velocity, which is not stiff because it vanishes for a homogeneous plasma). Then, the approximate Jacobian system reads:

$$J\delta\mathbf{x}\approx\begin{bmatrix} M & U \\ L & D_{\mathbf{v}}\end{bmatrix}\begin{pmatrix}\delta\mathbf{y}\\ \delta\mathbf{v}\end{pmatrix},$$

where $\delta\mathbf{y}=(\delta\rho,\delta\mathbf{A},\delta p)^T$, and

$$M\approx\begin{pmatrix} D_\rho & 0 & 0\\ L_{p\rho} & D_p & 0\\ L_{\mathbf{A}\rho} & L_{\mathbf{A}p} & D_{\mathbf{A}}\end{pmatrix} \tag{15}$$

a lower triangular system. The exact Schur factorization of the inverse of the 2×2 block Jacobian matrix yields:

$$\begin{bmatrix} M & U \\ L & D_{\mathbf{v}}\end{bmatrix}^{-1}=\begin{bmatrix} \mathbb{I} & -M^{-1}U\\ 0 & \mathbb{I}\end{bmatrix}\begin{bmatrix} M^{-1} & 0\\ 0 & P_{Schur}^{-1}\end{bmatrix}\begin{bmatrix} \mathbb{I} & 0\\ -LM^{-1} & \mathbb{I}\end{bmatrix},$$

where $P_{Schur}=D_{\mathbf{v}}-LM^{-1}U$ is the Schur complement, which contains all the information from the off-diagonal blocks L and U (similarly to the simple coupled-wave-equation example discussed earlier). This gives the *exact* inversion process:

$$\begin{aligned}\delta\mathbf{y}^* &= -M^{-1}\mathbf{G}_{\mathbf{y}},\\ \delta\mathbf{v} &= P_{Schur}^{-1}[-\mathbf{G}_{\mathbf{v}}-L\delta\mathbf{y}^*],\\ \delta\mathbf{y} &= \delta\mathbf{y}^*-M^{-1}U\delta\mathbf{v}.\end{aligned} \tag{16}$$

As it stands, however, inverting P_{Schur} is impractical due to the presence of M^{-1}. Suitable simplifications have been proposed for small ion flows [20], and later generalized to arbitrary ion flows [11], which cannot yet be directly applied in this context owing to the presence of the very stiff vector-potential diagonal block, $D_{\mathbf{A}}$. However, progress can be made by realizing that the vector-potential diagonal block is the only one supporting stiff EMHD physics, while the rest of the algorithm in Eq. 16 is correcting for ion dynamics. Since Hall physics decouples electrons from ions, we hypothesize that an operator-split approach in which the electrons response is solved for first, followed by an ion response correction will form the basis for an effective preconditioner.

Making this approximation explicit in Eq. 16, we perform a full solve of M^{-1} in the first step, while in subsequent ones we neglect electron dynamics and approximate $M^{-1}\approx\Delta t\,\mathbb{I}$ (small-bulk-flow approximation [20]). The resulting approximate "physics-based" preconditioner reads:

$$\begin{aligned}\delta\mathbf{y}^* &= -M^{-1}\mathbf{G}_{\mathbf{y}},\\ \delta\mathbf{v} &= P_{SF}^{-1}[-\mathbf{G}_{\mathbf{v}}-L\delta\mathbf{y}^*],\\ \delta\mathbf{y} &= \delta\mathbf{y}^*-\Delta t\,U\delta\mathbf{v},\end{aligned} \tag{17}$$

with $P_{SF}=D_{\mathbf{v}}-\Delta t\,LU$. This is formally identical to the resistive MHD preconditioner proposed in [20], except now the inversion of the M block contains the EMHD block, which is the stiffest in the system and

whose treatment we discuss in detail below. As in the reference, taking L and U from the linearized form of Eqs. 1-4, using the non-conservative form of Eq. 3, and Picard-linearizing the density (by ignoring $\delta\rho$ terms), the operator P_{SF} acting on $\delta\mathbf{v}$ reads:

$$P_{SF}\delta\mathbf{v} = \left[\rho_0\frac{\delta\mathbf{v}}{\Delta t} + \theta\rho_0(\mathbf{v}_0\cdot\nabla\delta\mathbf{v} + \delta\mathbf{v}\cdot\nabla\mathbf{v}_0) - \theta\nabla\cdot(\rho_0\nu_i\nabla\delta\mathbf{v})\right] + \Delta t\,\theta^2 W(\mathbf{B}_0, p_0)\delta\mathbf{v}, \tag{18}$$

where $W(\mathbf{B}_0, p_0)$ results from the composition of off-diagonal operators $L_{\mathbf{Av}}$ and $U_{\mathbf{vA}}$, and reads:

$$W(\mathbf{B}_0, p_0)\delta\mathbf{v} \approx \mathbf{B}_0\times\nabla\times\nabla\times[\delta\mathbf{v}\times\mathbf{B}_0] - \mathbf{j}_0\times\nabla\times[\delta\mathbf{v}\times\mathbf{B}_0] - \nabla[\delta\mathbf{v}\cdot\nabla p_0 + \gamma p_0\nabla\cdot\delta\mathbf{v}], \tag{19}$$

with $\mathbf{j}_0 = \nabla\times\mathbf{B}_0$. This is a linearized form of the MHD energy principle, which is self-adjoint when $\mathbf{j}_0\times\mathbf{B}_0 = \nabla p_0$ (and therefore has real eigenvalues in that case). Thus, the MHD system has been effectively parabolized.

Equations 17, with M defined in Eq. 15 and P_{SF} defined in Eq. 18, is the proposed preconditioner. It reuses many of the elements of the resistive MHD preconditioner proposed in Ref. [20], except for the treatment of the vector potential diagonal block, which we discuss next.

4. Numerical implementation

We employ a Newton-FGMRES (flexible GMRES [25]) nonlinear iterative method, implemented Jacobian-free. Newton nonlinear convergence is controlled in our implementation by the usual criterion:

$$\|\mathbf{G}(\mathbf{x}_k)\|_2 < \varepsilon_a + \varepsilon_r\|\mathbf{G}(\mathbf{x}_0)\|_2 = \varepsilon_t, \tag{20}$$

where $\|\cdot\|_2$ is the L_2-norm (euclidean norm), $\varepsilon_a = \sqrt{N}\times 10^{-15}$ (with N the total number of degrees of freedom) is an absolute tolerance to avoid converging below round-off, ε_r is the Newton relative convergence tolerance (set to 10^{-3} in this work), and $\mathbf{G}(\mathbf{x}_0)$ is the initial residual. GMRES convergence is controlled by an inexact Newton method [48] in which the Krylov convergence tolerance every Newton iteration is adjusted as follows:

$$\|J_k\delta\mathbf{x}_k + \mathbf{G}(\mathbf{x}_k)\|_2 < \zeta_k\|\mathbf{G}(\mathbf{x}_k)\|_2, \tag{21}$$

where ζ_k is the inexact Newton parameter and $J_k = \left.\frac{\partial\mathbf{G}}{\partial\mathbf{x}}\right|_k$ is the Jacobian matrix. Here, we employ the same prescription for ζ_k as in earlier studies [6, 20]:

$$\begin{aligned}
\zeta_k^A &= \gamma\left(\frac{\|\mathbf{G}(\mathbf{x}_k)\|_2}{\|\mathbf{G}(\mathbf{x}_{k-1})\|_2}\right)^{\xi},\\
\zeta_k^B &= \min[\zeta_{max}, \max(\zeta_k^A, \gamma\zeta_{k-1}^{\xi})],\\
\zeta_k &= \min[\zeta_{max}, \max(\zeta_k^B, \gamma\frac{\varepsilon_t}{\|\mathbf{G}(\mathbf{x}_k)\|_2})],
\end{aligned}$$

with $\xi = 1.5$, $\gamma = 0.9$, and $\zeta_{max} = 0.8$. The convergence tolerance ε_t is the same as in Eq. 20. In this prescription, the first step ensures superlinear convergence (for $\zeta > 1$), the second avoids volatile decreases in ζ_k, and the last avoids oversolving in the last Newton iteration.

Our geometric MG solver is fully parallel, with coarse-grid parallel domain agglomeration [26, 27]. It features a matrix-light implementation [6, 20], in which only the diagonal of the system of interest is stored for smoothing purposes. Coarse operators are found via rediscretization, and required residuals in the MG iteration are found in a matrix-free manner. As a smoother, we employ three passes of damped Jacobi (p. 10 in [49]; p. 118 in [36]), with damping weight $\sigma = 0.7$, for both the pre- and post-smoothing steps. MG restriction employs conservative agglomeration, and prolongation employs a first-order interpolation.

In MG jargon, such V-cycle is identified as V(3,3), where the two integers indicate restriction and prolongation smoothing steps, respectively. In parallel, we use colored damped Jacobi, which parallelizes very well; four colors are needed in 2D (for a 9-point stencil), and 8 colors in 3D (for a 27-point stencil). Our matrix-light geometric-MG approach features excellent parallel performance, with wall-clock time scaling logarithmically with the number of MPI tasks (as demonstrated in the next section).

For typical timesteps in HMHD, a single V(3,3) MG V-cycle is sufficient to approximately invert the diagonal blocks D_ρ, D_p, and P_{SF} in Eq. 17. However, the EMHD block $D_{\mathbf{A}}$ is much more challenging to invert robustly. For this block, we use MG-preconditioned GMRES, converged to either 10 Krylov iterations or a relative tolerance of 0.1, whichever comes first (note that nested GMRES solves in the preconditioner are allowed by FGMRES). This strategy still requires a working MG solver for the EMHD block, which in turn requires a suitable smoother. We describe our smoothing strategy for this block next.

4.1. Smoothing of the vector-potential diagonal block

In order to make the $D_{\mathbf{A}}$ operator MG-friendly we need to provide a suitable smoother. Here, we are seeking to use damped block-Jacobi smoothing, which is cheap and easily parallelizable. The $D_{\mathbf{A}}$ diagonal block contains fast electron hyperbolic physics (whistler waves, with dispersion relation $\omega \sim d_i v_A k k_\parallel$), electron inertia, and electron viscosity (also known as hyperresistivity), and reads:

$$\begin{aligned} D_{\mathbf{A}}\delta\mathbf{A} &= \frac{\delta\mathbf{A}^*}{\Delta t} \\ &- \theta\left(\mathbf{v}_{e,0}\times\nabla\times\delta\mathbf{A}^* - d_i\underbrace{\frac{\nabla\times\nabla\times\delta\mathbf{A}}{\rho_0}\times\mathbf{B_0}}_{\text{Whistler term}} - \eta\nabla\times\nabla\times\delta\mathbf{A} + \underbrace{\nabla\cdot\left[\rho_0 d_i^2\nu_e\nabla\left(\frac{\nabla\times\nabla\times\delta\mathbf{A}}{\rho_0}\right)\right]}_{\text{Hyperresistivity term}}\right), \end{aligned}$$

with:

$$\delta\mathbf{A}^* = \delta\mathbf{A} + \frac{d_e^2}{\rho_0}\nabla\times\nabla\times\delta\mathbf{A}.$$

This operator is the linearized form of the EMHD model.

The iterative solution of the vector-potential diagonal block is extremely challenging, and has in fact been the key roadblock overcome in this research. It is a stiff, high differential order, vectorial system of equations. It supports fast (quadratic) dispersive waves, and features a fourth-order dissipative operator of the form $\nabla^2(\nabla\times\nabla\times)$ that provides a dissipation length scale to the fast dispersive waves, needed for nonlinear stability of the discretization [2]. The form of the dissipative operator is not only problematic because of its high order, but also because it features an infinitely degenerate null space: any gradient component in the vector potential will get annihilated by the curl. Although the $D_{\mathbf{A}}$ operator in aggregate is non-singular, the presence of a null space in the hyperresistivity term makes its MG implementation problematic. One may consider approximating it, but being the term with the highest differential order, our experience is that any approximations introduced to it will negatively (and significantly) impact the performance of the preconditioner. In particular, replacing $\nabla\times\nabla\times$ by $-\nabla^2$ is not justified in principle because we do not use the Coulomb gauge ($\nabla\cdot\mathbf{A}=0$) but a Weyl gauge, as discussed in Appendix B. (A Coulomb gauge could in principle be used, but it would result in an additional equation for the electrostatic potential, which the Weyl gauge avoids.)

To provide for a suitable smoother, we reformulate the EMHD system by splitting it into two second-order vectorial systems:

$$\frac{1}{\Delta t}\left(\delta\mathbf{A}+d_e^2\frac{\delta\mathbf{j}}{\rho_0}\right) - \theta\left(\mathbf{v}_{e,0}\times\nabla\times\left(\delta\mathbf{A}+d_e^2\frac{\delta\mathbf{j}}{\rho_0}\right) - d_i\frac{\delta\mathbf{j}}{\rho_0}\times\mathbf{B_0} - \eta\,\delta\mathbf{j} + \nabla\cdot\left[\rho_0 d_i^2\nu_e\nabla\left(\frac{\delta\mathbf{j}}{\rho_0}\right)\right]\right) = \mathbf{r}_{\mathbf{A}} \quad (22)$$

$$\delta\mathbf{j} - \nabla\times\nabla\times\delta\mathbf{A} = \mathbf{0} \quad (23)$$

The auxiliary variable $\delta\mathbf{j}$ is a physical quantity (current density), with a well-defined set of boundary conditions, ensuring the split system is well posed. The feasibility of this physical system splitting without approximation is, in fact, the key reason why we pursued a vector-potential-based representation of our Hall MHD model instead of a magnetic-field-based one. For the latter, the hyperresistivity term would read:

$$\nabla\times\nabla\cdot\left[\rho_0 d_i^2\nu_e\nabla\left(\frac{\nabla\times\delta\mathbf{B}}{\rho_0}\right)\right],$$

which *cannot* be split into two second-order systems in an obvious way.

The Jacobi smoother for the coupled system in Eqs. 22, 23 iterates on all local vector components for $\delta\mathbf{A}$ and $\delta\mathbf{j}$ simultaneously on a per-cell basis. To deal with the $\nabla\times\nabla\times$ operator in the smoothing stage, we employ a defect-correction approach whereby we selectively approximate $\nabla\times\nabla\times$ by $-\nabla^2$ in the computation of the diagonal blocks, but retain the $\nabla\times\nabla\times$ form for all the residual computations within the MG V-cycle. We show in Appendix C that, for the special case of a homogeneous plasma, this strategy produces a convergent damped-Jacobi iteration with the smoothing property (critical for the effectiveness of MG), and therefore provides an excellent basis for an effective MG preconditioner.

5. Numerical results

We have implemented this algorithm in the PIXIE3D nonlinear MHD code [33, 20, 11]. In what follows, we present numerical examples that demonstrate the correctness of our implementation as well as its performance (algorithmically and in parallel). We consider both linear (wave propagation) tests and nonlinear tests such as the well-known GEM challenge magnetic reconnection problem [2], which we verify against the HiFi nonlinear MHD code [50]. The latter example is also considered for the algorithmic and parallel performance assessment. The GEM challenge problem benefits from an implicit treatment, since the dynamical time scale of interest is independent of the grid resolution, and much slower than fast HMHD normal modes. In what follows, we neglect electron inertia (i.e., $d_e = m_e = 0$). Even though electron inertia is known to arrest the fast dispersive frequency growth in HMHD [51] (and therefore improves numerical conditioning), it is ignored in this study because 1) it does not prevent nonlinear singular behavior in HMHD [52] (and therefore electron viscosity is still needed for regularity), and 2) removing it will allow us to stress-test our preconditioner strategy in the fast-dispersive regime as we increase resolution without interference.

5.1. Linear verification: whistler wave

We consider a collisionless, homogeneous, adiabatic ($\gamma = 1$) plasma with $\rho_0 = 1$, $T_{i0} = T_{e0} = 1$, $\mathbf{B}_0 = (1,0,0)$, $\mathbf{v} = \mathbf{0}$, and $d_i = 10$ in a one-dimensional domain with $L_x = 1$. Linear theory predicts that the frequency of the resulting wave is $\omega = \pm d_i k_x^2$, with mode structure:

$$\begin{aligned}\delta A_y &= -i\delta A_z,\\ \omega\delta v_y &= k_x^2\delta A_y,\\ \omega\delta v_z &= k_x^2\delta A_z.\end{aligned}$$

For $k_x = 2\pi/L_x = 2\pi$, the wave period is $T = 2\pi/\omega = (2\pi d_i)^{-1} \approx 0.016$. We set up a standing whistler wave with a perturbation of the form:

$$\begin{aligned}\delta A_z &= -\frac{\varepsilon}{k_x}\sin(k_x x),\\ \delta v_z &= \frac{\varepsilon}{k_x d_i}\sin(k_x x),\end{aligned}$$

with $\varepsilon = 10^{-3}$. Simulations are performed using 64 mesh points and $\Delta t = 10^{-4}$ with a Crank-Nicolson temporal scheme, which does not introduce any temporal dissipation. Figure 1 shows the time histories of the ℓ_2-norm of δv_y, δv_z, δA_y, δA_z, along with the analytical expectations, demonstrating excellent agreement.

5.2. Linear verification: kinetic Alfvén wave (KAW)

We consider a collisionless, homogeneous, adiabatic ($\gamma = 1$) plasma with $\rho_0 = 1$, $T_{i0} = T_{e0} = 2.5 \times 10^3$, $\mathbf{B}_0 = (1,0,0)$, $\mathbf{v} = \mathbf{0}$, and $d_i = 5/\pi$ in a two-dimensional domain with $L_x = 10$ and $L_y = 1$. Linear theory predicts that the frequency of the resulting wave is $\omega \approx \pm \rho_s k_x k_y$, with $\rho_s = d_i \sqrt{\beta_p/2}$ the sound Larmor radius and $\beta_p = 2\mu_0 p_0 / B_0^2 = 10^4$ the plasma beta, and mode structure:

$$\delta A_z = \frac{i}{\rho_s k_x k_y} \delta v_y, \tag{24}$$

$$\delta v_y = \frac{k_x}{k_y} [d_i^2 (k_x^2 + k_y^2) - 1] \delta v_x. \tag{25}$$

For $k_x = 2\pi/L_x = \pi/5$, $k_y = 2\pi n_y / L_y = 2\pi n_y$, and $n_y = 5$, the wave period is $T = 2\pi/\omega = n_y^{-1} \sqrt{2/\beta_p} \approx 2.83 \times 10^{-3}$. We set up a standing wave with a perturbation of the form:

$$\begin{aligned} \delta v_x &= \varepsilon \cos(k_x x + k_y y), \\ \delta v_y &= \varepsilon \frac{k_x}{k_y} [d_i^2 (k_x^2 + k_y^2) - 1] \cos(k_x x + k_y y), \end{aligned}$$

with $\varepsilon = 10^{-7}$. Simulations are performed using 64×256 mesh points and $\Delta t = 4 \times 10^{-5}$, also with Crank-Nicolson. Figure 2 shows the time histories of the ℓ_2-norm of δv_x, δv_y, δA_z, along with analytical expectations, again demonstrating excellent agreement.

5.3. Nonlinear verification: GEM challenge magnetic reconnection problem

We consider the GEM challenge problem [2] for a nonlinear verification study against the HiFi nonlinear extended MHD code. HiFi [50] is a highly accurate, fully implicit, spectral-element-based nonlinear extended MHD code. To preserve smoothness during the simulation, HiFi features a moving mesh strategy based on a Laplace-Beltrami mesh governing equation (see Ref. [54] and references therein). For consistency with HiFi, we have replaced $\mathbf{v}^*$ in Eq. 4 with $\mathbf{v}$.

The GEM challenge problem setup is a Harris current sheet, defined by the vector potential $(0,0,A_z)$, with:

$$A_z = -\lambda \ln(\cosh(x/\lambda)),$$

with $\lambda = 0.5\, d_i$, balanced by a pressure profile given by constant ion and electron temperatures with $T_i/T_e = \alpha = 5$ and $(T_i + T_e) = 1/2$, and density profile:

$$\rho(x) = \frac{n_0}{\cosh(x/\lambda)} + n_\infty.$$

with $n_0 = 1$, $n_\infty = 0.2$. We consider a slab domain $(-L_x/2, L_x/2) \times (-L_y/2, L_y/2)$ with $L_x = 12.8\, d_i$ and $L_y = 25.6\, d_i$. Boundary conditions are perfect conductor at $\pm L_x/2$, and periodic at $\pm L_y/2$. The transport coefficients are: $\nu_i = 5 \times 10^{-2}$, $\nu_e = 10^{-4}$, $\eta = 5 \times 10^{-3}$, and $\kappa = 2 \times 10^{-2}$. The specific heat ratio is $\gamma = 5/3$.

The simulation in both codes is started by perturbing A_z with a sinusoidal perturbation of the form:

$$\delta A_z = -\varepsilon \cos\left(\frac{\pi x}{L_x}\right) \cos\left(\frac{2\pi y}{L_y}\right),$$

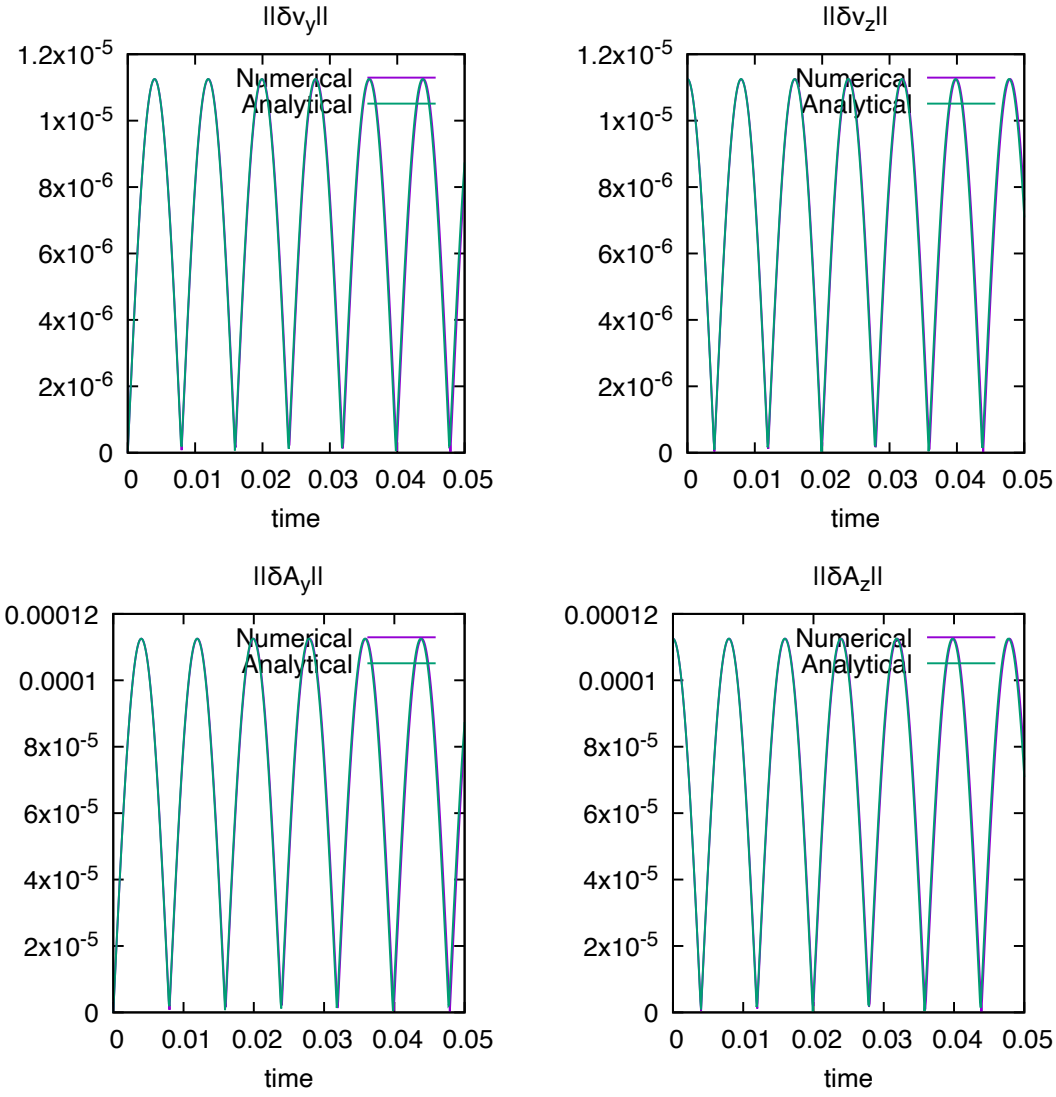


Figure 1: Whistler wave test: Time histories of perturbations of v_y, v_z, A_y, A_z for the whistler wave linear verification test, along with comparisons with their expected analytical evolution.

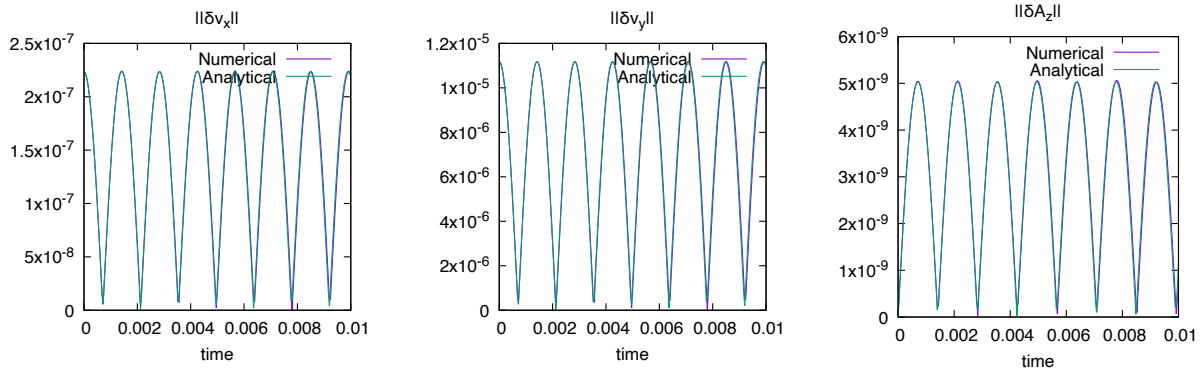


Figure 2: KAW test: Time histories of perturbations of v_x, v_y, A_z for the KAW wave linear verification test, along with comparisons with their expected analytical evolution.

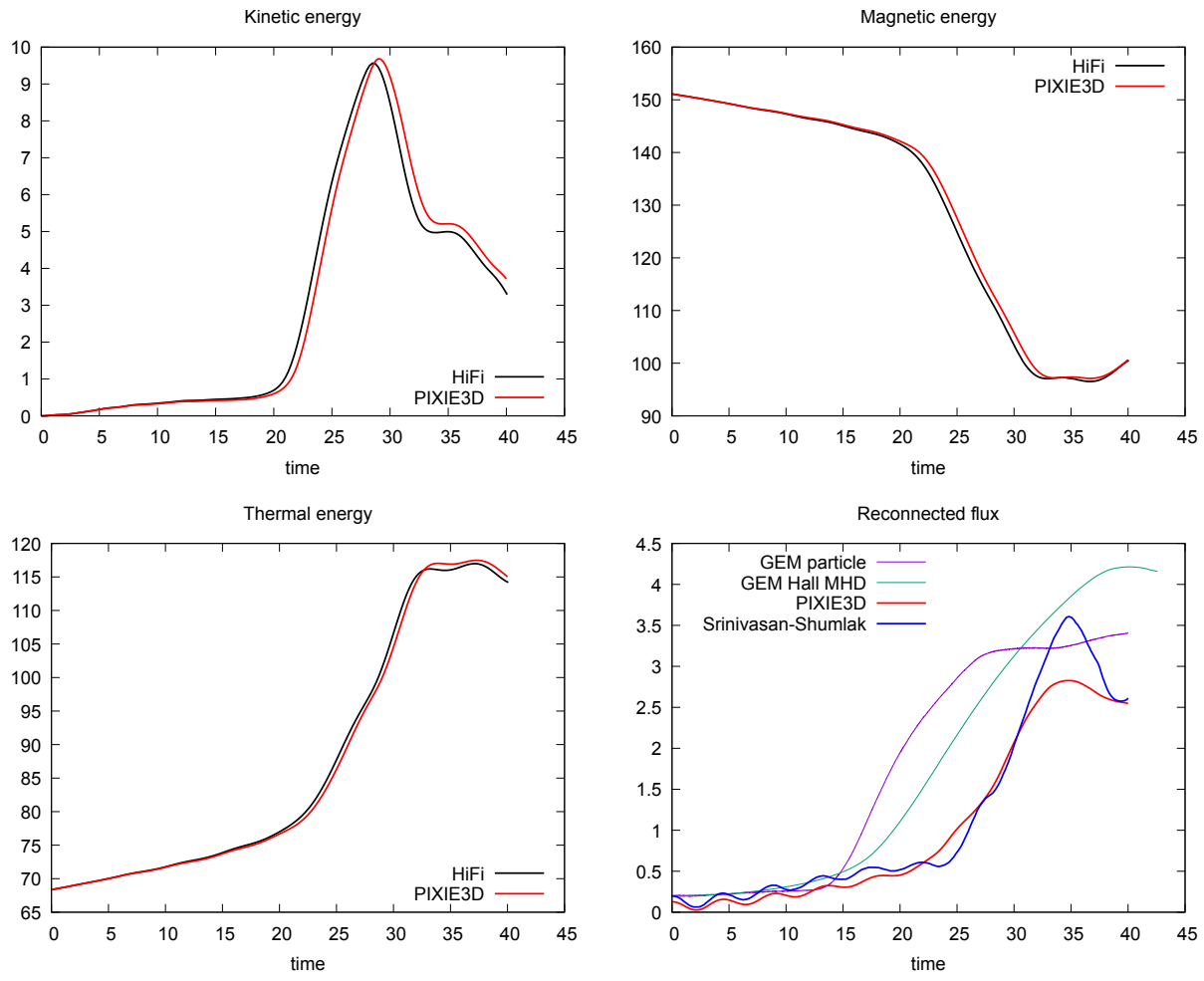


Figure 3: GEM challenge test: Nonlinear comparison of the temporal evolution of the kinetic, magnetic and thermal energies from the PIXIE3D and HiFi codes for the GEM challenge (see text for setup). The reconnected flux for PIXIE3D is compared with data from Fig. 1 of the GEM-challenge study [2] and from Fig. 13 of Ref. [53], with excellent agreement in the slope of the curve (which determines the peak reconnection rate) vs. the GEM results and in quantitative agreement with Ref. [53].

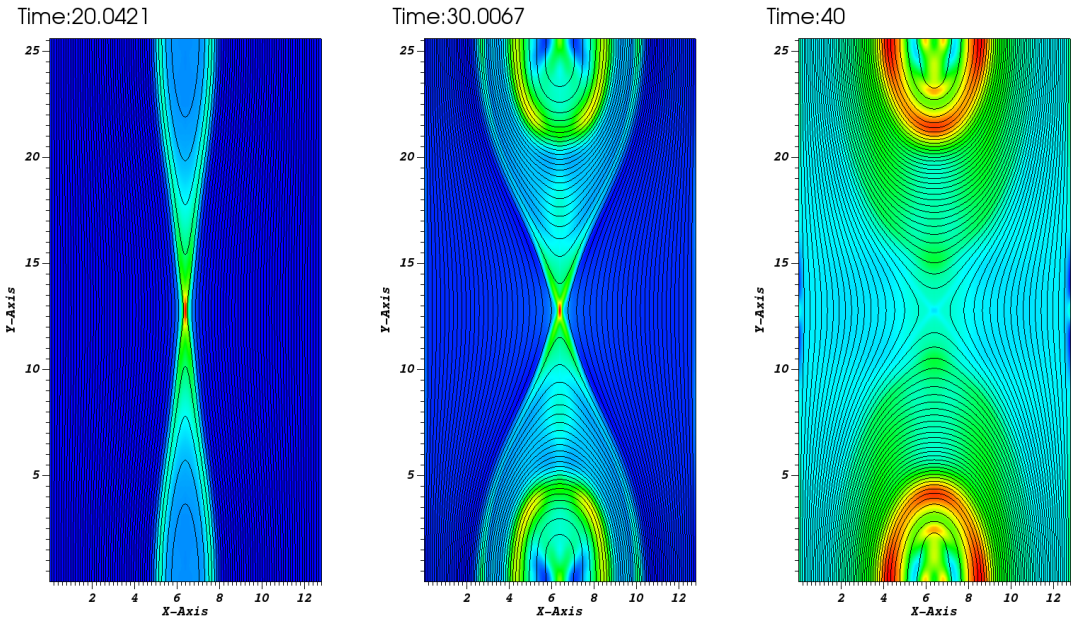


Figure 4: Snapshots of A_z (contours) and j_z (color) at three different times ($t = 20, 30, 40$) for the GEM simulation.

with $\varepsilon = 10^{-1}$. This problem was run with a 256×256 packed mesh along both axes, with $\Delta x = 2 \times 10^{-2}$ at $x = 0$ (at the rational surface), and $\Delta y = 5 \times 10^{-2}$ at $y = \pm L_y/2$ (the periodic boundaries). This represents a packing ratio vs. the uniform mesh of 2.5 in x and 2 in y. The time step is $\Delta t = 0.05$, and we use the BDF2 temporal scheme.

In addition to the reconnected flux (which is a robust measure [2]), we also track the temporal histories of kinetic ($\frac{1}{2}\int d\mathbf{x}\rho v^2$), magnetic ($\frac{1}{2}\int d\mathbf{x}B^2$), and thermal ($\frac{1}{2}\int d\mathbf{x}\frac{p}{\gamma-1}$) energies, which are more nuanced. The time history comparison is depicted in Fig. 3, and shows an early (slow) phase followed by a late, explosive phase characterized by rapid growth of kinetic energy and thermal energy, fed by a significant decrease in magnetic energy. During this evolution, total energy is conserved better than 0.04%. The comparison shows remarkably good agreement between HiFi and PIXIE3D during the slow phase and most of the explosive phase, with some deviation (of about 1%) late in the nonlinear evolution of the problem. Key elements of comparison are the excellent agreement in the timing of the transition from slow to explosive phase, in the slopes during most of the explosive phase, and in the extrema values of the energies at saturation. The reconnected flux for PIXIE3D is also shown, along with fully kinetic and Hall MHD simulation data from the GEM-challenge study [2], and from Fig. 13 of Ref. [53]. While details differ in terms of timing and saturation level, the agreement in the slope (maximum reconnection rate) between all four simulations is apparent. It is also apparent that the PIXIE3D reconnected-flux history is in excellent quantitative agreement with the more recent HMHD GEM results of Ref. [53].

Snapshots of the z-component of the vector potential and the current density for the GEM challenge at $t = 20$ (before the explosive phase), $t = 30$ (at the peak of the explosive phase), and $t = 40$ (at the end of the simulation) are depicted in Fig. 4. The formation of an open X-point and associated current density spike is apparent. We do not find the formation of a central magnetic island for our particular choice of parameters, which has been the subject of contention in the literature [13, 19].

5.4. Parallel and algorithmic performance

We assess the (weak) parallel and algorithmic scalability of our solver with the GEM challenge problem, with slightly modified parameters as indicated below. Preconditioner setup and linear and nonlinear convergence tolerances are set as stated earlier. For this exercise, we consider uniform meshes in full domain

Table 1: Parallel weak-scaling performance for the GEM Challenge problem with $\Delta t = 0.01$ and $\eta = \nu_i = \kappa = 5 \times 10^{-4}$. Results are obtained in full domain and a uniform Cartesian mesh, and averaged over 100 time steps with a fixed timestep $\Delta t = 0.01$. In this table, WCT stands for wall-clock time. Speedup is vs. a predictor-corrector explicit method. The average number of GMRES and Newton iterations per timestep are the same because the preconditioner results in a single GMRES iteration per Newton iteration.

n_p	mesh	ν_e	$\frac{\Delta t}{\Delta t_{exp}}$	GMRES$/\Delta t$	Newton$/\Delta t$	WCT(s)	WCT/GMRES	Speedup
1	32×32	10^{-4}	1.0	2.0	2.0	132	66	0.04
4	64×64	2.5×10^{-5}	2.1	2.6	2.6	172	66	0.07
16	128×128	6.3×10^{-6}	8.3	2.0	2.0	182	105	0.26
64	256×256	1.6×10^{-6}	33	2.0	2.0	303	151	0.63
256	512×512	3.9×10^{-7}	130	2.0	2.0	321	160	3.0
1024	1024×1024	9.8×10^{-8}	520	2.0	2.0	509	203	8.5
4096	2048×2048	2.4×10^{-8}	2100	2.0	2.0	512	256	38
16384	4096×4096	6.0×10^{-9}	8400	3.0	3.0	1130	377	75

starting at 32×32 with $n_p = 1$ MPI task, up to 4096×4096 meshes with $n_p = 16384$ tasks. We consider a fixed implicit timestep ($\Delta t = 0.01$) for all experiments, so the problem is becoming numerically stiffer as the number of MPI tasks grows. In this way, when increasing the processor count we are able to test the ability of the algorithm to *both* scale with mesh refinement and to scale in parallel. To maintain the simulation in the stiff hyperbolic regime, we decrease the hyperresistivity coefficient as we refine the mesh according to Eq. 10 to ensure that high-order dissipation is only provided for the grid-scale components of the solution.

The results of the mesh/processor convergence study are shown in Table 1. Timings have been obtained on the CPU partition of the Chicoma computer at Los Alamos National Laboratory, featuring 1792 nodes, each with two Rome 64 core 2.6 GHz CPUs (also called the AMD EPYC 7H12) with 512 GB per node. Several observations are in order. Firstly, as advanced earlier, the fixed implicit timestep results in a stiffer numerical setup as we refine the mesh, and indicated by the increase in $\Delta t/\Delta t_{exp}$ (with Δt_{exp} the explicit stability-limit CFL timestep). Next, the total number of GMRES iterations per timestep (GMRES/Δt, which includes all Newton iterations) shows a very weak dependence with mesh resolution (and number of MPI tasks), attesting to the effectiveness of our preconditioning strategy. The performance of our MG parallel implementation is demonstrated by the WCT/GMRES column, which measures the time per GMRES iteration, which is dominated by the MG preconditioning step and features a $log(n_p)$ increase. The favorable algorithmic (per the GMRES/Δt column) and parallel (per the WCT/GMRES column) scalings result in an overall remarkable scaling of the wall-clock time WCT, which only increases by a factor of 8.6 when increasing the problem size by a factor of 16384. It also results in a speedup of 75 for the finest mesh considered vs. an explicit integrator, with performance cross-over between implicit and explicit happening at moderate meshes (between 256×256 and 512×512).

6. Conclusions

We have developed a parallel, scalable multigrid-based physics-based preconditioning strategy for the Hall MHD equation system. As in earlier MHD preconditioning studies, the physics-based preconditioning approach exploits the natural parabolization of the hyperbolic couplings that occurs when one discretizes them implicitly. The block structure of the Hall MHD Jacobian spoils the "arrow" structure that was so convenient in resistive MHD, but can still be formally Schur-factorized in a similar way leading to a coupled system of equations for the fluid velocity components. The Schur factorization is rendered practical by 1) using the small-bulk-flow approximation (as in earlier studies), and 2) exploiting the natural separation of stiff timescales between ions and electrons in Hall MHD to segregate the electromagnetic diagonal block solve.

The electromagnetic diagonal block corresponds to the so-called electron MHD model, and supports fast (quadratic) dispersive waves and high-order dissipation (hyperresistivity), which is critical for nonlinear stability. We invert this system robustly and scalably with MG-preconditioned GMRES (nested GMRES solves in the preconditioning stage are allowed by FGMRES in JFNK). Several strategies have helped in rendering the EMHD block MG-friendly. Firstly, we have chosen to solve the vector-potential formulation of Hall MHD (instead of the more standard magnetic field one), which allowed us to reformulate the EMHD block into a set of two coupled vectorial equations with physical interpretation (vector potential and current density, respectively) and well-defined boundary conditions. This coupled set of equations is Jacobi-smoothed in a fully coupled manner. Secondly, we use a deferred-correction approach in our MG smoothing strategy, whereby we approximate the $\nabla \times \nabla \times$ operator (which relates the current density and the vector potential) as a vector Laplacian for the construction of the Jacobi block diagonals. This strategy is demonstrated by analysis to produce a damped Jacobi iteration with the smoothing property, and therefore a robust MG algorithm.

We have verified our implementation linearly with two wave propagation tests, a whistler wave and a KAW, and nonlinearly with the GEM challenge problem, by direct comparison with the HiFi spectral extended MHD code. We have demonstrated the excellent scalability properties of our preconditioner algorithmically (by demonstrating a bounded iteration count with grid refinement) and in parallel (by demonstrating logarithmic scaling of wall clock time with the number of MPI tasks in a weak scaling sense with up to 16384 processors).

Future work will focus on extending the algorithm to polar coordinate systems (with application to cylindrical and toroidal geometries, of interest for fusion plasma simulation), and to include separate ion and electron temperatures and gyroviscosity.

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Appendix A. Derivation of the energy conservation theorem for the HMHD model in Eqs. 1-5

We derive here the total energy conservation equation, Eq. 6. We begin with the HMHD model, which we reproduce here for completeness:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \tag{A.1}$$

$$\frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} = 0 \tag{A.2}$$

$$\frac{\partial (\rho \mathbf{v})}{\partial t} + \nabla \cdot \left[\rho \mathbf{v}\mathbf{v} - \mathbf{B}\mathbf{B} + \mathbb{I}(p + \frac{B^2}{2}) + \sum_{s=i,e} \Pi_s \right] = 0, \tag{A.3}$$

$$\frac{\partial p}{\partial t} + \nabla \cdot (\mathbf{v}^* p) + (\gamma - 1) p \nabla \cdot \mathbf{v}^* = (\gamma - 1)(\eta j^2 - \sum_{s=i,e} \Pi_s : \nabla \mathbf{v}_s - \nabla \cdot \mathbf{q}), \tag{A.4}$$

$$\mathbf{E} = -\mathbf{v} \times \mathbf{B} + \eta \mathbf{j} + \frac{d_i}{\rho}(\mathbf{j} \times \mathbf{B} - \nabla p_e - \nabla \cdot \Pi_e) + d_e^2 \frac{d}{dt}\left(\frac{\mathbf{j}}{\rho}\right) \tag{A.5}$$

We begin by Eq. A.3 with $\mathbf{v}$, to find:

$$\frac{\partial (\rho v^2/2)}{\partial t} + \nabla \cdot \left[\rho \mathbf{v} \frac{v^2}{2} \right] + (\nabla \cdot \sum_{s=i,e} \Pi_s) \cdot \mathbf{v} = \mathbf{j} \cdot (-\mathbf{v} \times \mathbf{B}) - \mathbf{v} \cdot \nabla p. \tag{A.6}$$

Using Eq. A.5, we find:

$$\mathbf{j} \cdot (-\mathbf{v} \times \mathbf{B}) = \mathbf{E} \cdot \mathbf{j} - \eta j^2 + \frac{d_i}{\rho}(\mathbf{j} \cdot \nabla p_e + \mathbf{j} \cdot \nabla \cdot \Pi_e) - d_e^2 \mathbf{j} \cdot \frac{d}{dt}\left(\frac{\mathbf{j}}{\rho}\right). \tag{A.7}$$

Now dotting Eq. A.2 with $\mathbf{B}$, we find:

$$\frac{\partial B^2/2}{\partial t} + \nabla \cdot (\mathbf{E} \times \mathbf{B}) + \mathbf{E} \cdot \mathbf{j} = 0,$$

which when introduced in Eq. A.7 together with:

$$\frac{d_i}{\rho}\mathbf{j} = \mathbf{v} - \mathbf{v}_e, \tag{A.8}$$

gives:

$$\mathbf{j} \cdot (-\mathbf{v} \times \mathbf{B}) = -\frac{\partial B^2/2}{\partial t} - \nabla \cdot (\mathbf{E} \times \mathbf{B}) - \eta j^2 - (\mathbf{v}_e - \mathbf{v}) \cdot \nabla p_e - (\mathbf{v}_e - \mathbf{v}) \cdot \nabla \cdot \Pi_e - d_e^2 \mathbf{j} \cdot \frac{d}{dt}\left(\frac{\mathbf{j}}{\rho}\right).$$

Introducing this equation into Eq. A.6 gives:

$$\frac{\partial}{\partial t}\left(\rho \frac{v^2}{2} + \frac{B^2}{2}\right) + \nabla \cdot \left[\rho \mathbf{v} \frac{v^2}{2} + \mathbf{E} \times \mathbf{B}\right] = -\eta j^2 - \sum_{s=i,e} \mathbf{v}_s \cdot \nabla p_s - \sum_{s=i,e} \mathbf{v}_s \cdot \nabla \cdot \Pi_s - d_e^2 \mathbf{j} \cdot \frac{d}{dt}\left(\frac{\mathbf{j}}{\rho}\right). \tag{A.9}$$

Next, we rewrite Eq. A.4 in terms of the specific internal energy $e = p/(\gamma - 1)$ as:

$$\frac{\partial e}{\partial t} + \nabla \cdot (\mathbf{v}^* e) + p \nabla \cdot \mathbf{v}^* = (\eta j^2 - \sum_{s=i,e} \Pi_s : \nabla \mathbf{v}_s - \nabla \cdot \mathbf{q}),$$

and we add it to Eq. A.9 to find:

$$\frac{\partial}{\partial t}\left(\rho\frac{v^2}{2}+\frac{B^2}{2}+e\right)+\nabla\cdot\left[\rho\mathbf{v}\frac{v^2}{2}+\mathbf{E}\times\mathbf{B}+\mathbf{q}+\sum_{s=i,e}\mathbf{v}_s\cdot\Pi_s\right]+\nabla\cdot(\mathbf{v}^*e)+p\nabla\cdot\mathbf{v}^*=-\sum_{s=i,e}\mathbf{v}_s\cdot\nabla p_s-d_e^2\mathbf{j}\cdot\frac{d}{dt}\left(\frac{\mathbf{j}}{\rho}\right).$$

Noting that:

$$v^*=\frac{\alpha\mathbf{v}+\mathbf{v}_e}{1+\alpha}\ ;\ p=(1+\alpha)p_e\ ;\ p_i=\alpha p_e,$$

we find:

$$\begin{aligned}p\nabla\cdot\mathbf{v}^*+\nabla\cdot(\mathbf{v}^*e)+\sum_{s=i,e}\mathbf{v}_s\cdot\nabla p_s &= p_e\nabla\cdot(\alpha\mathbf{v}+\mathbf{v}_e)+\nabla\cdot\left((\alpha\mathbf{v}+\mathbf{v}_e)\frac{p_e}{\gamma-1}\right)+\sum_{s=i,e}\mathbf{v}_s\cdot\nabla p_s\\ &= \frac{\gamma}{\gamma-1}\sum_{s=i,e}\nabla\cdot(\mathbf{v}_s p_s)=\gamma\nabla\cdot(e\mathbf{v}^*),\end{aligned}$$

and therefore the energy conservation equation reads:

$$\frac{\partial}{\partial t}\left(\rho\frac{v^2}{2}+\frac{B^2}{2}+e\right)+\nabla\cdot\left[\rho\mathbf{v}\frac{v^2}{2}+\mathbf{E}\times\mathbf{B}+\mathbf{q}+\sum_{s=i,e}\mathbf{v}_s\cdot\Pi_s+\gamma e\mathbf{v}^*\right]=-d_e^2\mathbf{j}\cdot\frac{d}{dt}\left(\frac{\mathbf{j}}{\rho}\right).$$

The last term in the right hand side can be written as:

$$d_e^2\mathbf{j}\cdot\frac{d}{dt}\left(\frac{\mathbf{j}}{\rho}\right)=\frac{d_e^2}{2}\rho\frac{d}{dt}\left(\frac{j}{\rho}\right)^2=\frac{d_e^2}{2}\rho\left[\partial_t\left(\frac{j}{\rho}\right)^2+\mathbf{v}_e\cdot\nabla\left(\frac{j}{\rho}\right)^2\right]=d_e^2\left[\partial_t\left(\frac{j^2}{2\rho}\right)+\nabla\cdot\left(\mathbf{v}_e\frac{j^2}{2\rho}\right)\right],$$

where in the last step we have used that, owing to $\nabla\cdot\mathbf{j}=0$, from Eqs. A.1 and A.8 we have:

$$\partial_t\rho+\nabla\cdot(\rho\mathbf{v}_e)=0.$$

We finally obtain:

$$\frac{\partial}{\partial t}\left(\rho\frac{v^2}{2}+\frac{B^2}{2}+e+d_e^2\frac{j^2}{2\rho}\right)+\nabla\cdot\left[\rho\mathbf{v}\frac{v^2}{2}+\mathbf{E}\times\mathbf{B}+\mathbf{q}+\sum_{s=i,e}\mathbf{v}_s\cdot\Pi_s+\gamma e\mathbf{v}^*+d_e^2\mathbf{v}_e\frac{j^2}{2\rho}\right]=0.$$

Appendix B. Derivation of vector potential evolution equation, Eq. 8

We begin with the vector potential evolution equation, Eq. 7:

$$\frac{\partial\mathbf{A}}{\partial t}+\mathbf{E}=\frac{\partial\mathbf{A}}{\partial t}-\mathbf{v}_e\times\mathbf{B}+\eta\mathbf{j}-\frac{d_i}{\rho}(\nabla p_e+\nabla\cdot\Pi_e)+d_e^2\frac{d}{dt}\left(\frac{\mathbf{j}}{\rho}\right)=-\nabla\Phi, \tag{B.1}$$

and write:

$$\rho\frac{d}{dt}\left(\frac{\mathbf{j}}{\rho}\right)=\rho\partial_t\left(\frac{\mathbf{j}}{\rho}\right)+\rho\mathbf{v}_e\cdot\nabla\left(\frac{\mathbf{j}}{\rho}\right)=\partial_t\mathbf{j}+\nabla\cdot(\mathbf{v}_e\mathbf{j}).$$

Choosing the Weyl Gauge $\Phi=0$, we find:

$$\frac{\partial\mathbf{A}}{\partial t}-\mathbf{v}_e\times(\nabla\times\mathbf{A})+\eta\nabla\times\nabla\times\mathbf{A}-\frac{d_i}{\rho}(\nabla p_e+\nabla\cdot\Pi_e)+\frac{d_e^2}{\rho}\left(\partial_t\mathbf{j}+\nabla\cdot(\mathbf{v}_e\mathbf{j})\right)=\mathbf{0}. \tag{B.2}$$

If, instead of the inertia term in Eq. A.5 (which results in an exact energy conservation theorem as indicated in the previous section), we use the more physically motivated one, $\frac{d_e^2}{d_i}\frac{d\mathbf{v}_e}{dt}$, such that the Ohm's law reads:

$$\mathbf{E} = -\mathbf{v}\times\mathbf{B} + \eta\mathbf{j} + \frac{d_i}{\rho}(\mathbf{j}\times\mathbf{B} - \nabla p_e - \nabla\cdot\Pi_e) - \frac{d_e^2}{d_i}\frac{d\mathbf{v}_e}{dt},$$

then, since:

$$\frac{d\mathbf{v}_e}{dt} = \partial_t\mathbf{v}_e + \mathbf{v}_e\cdot\nabla\mathbf{v}_e = \partial_t\mathbf{v}_e - \mathbf{v}_e\times\nabla\times\mathbf{v}_e + \nabla\frac{v_e^2}{2}.$$

we can write (choosing the Gauge $\Phi = -\frac{d_e^2}{d_i}\frac{v_e^2}{2}$):

$$\frac{\partial\mathbf{A}^*}{\partial t} - \mathbf{v}_e\times(\nabla\times\mathbf{A}^*) + \eta\nabla\times\nabla\times\mathbf{A} - \frac{d_i}{\rho}(\nabla p_e + \nabla\cdot\Pi_e) = \mathbf{0}, \tag{B.3}$$

with:

$$\mathbf{A}^* = \mathbf{A} - \frac{d_e^2}{d_i}\mathbf{v}_e.$$

This form is more straightforward to implement, but does not feature a strict local energy conservation law, featuring additional non-conservative terms of $\mathscr{O}(d_e^2)$. However, it is more convenient for preconditioning purposes than Eq. B.2. For that purpose, we consider a similar reformulation of Eq. B.2, which with a suitable Gauge can be rewritten as:

$$\frac{\partial\mathbf{A}^\dagger}{\partial t} - \mathbf{v}_e\times(\nabla\times\mathbf{A}^\dagger) + \eta\nabla\times\nabla\times\mathbf{A} - \frac{d_i}{\rho}(\nabla p_e + \nabla\cdot\Pi_e) = d_e^2\nabla\left(\frac{\mathbf{j}}{\rho}\right)\cdot\mathbf{v}, \tag{B.4}$$

with:

$$\mathbf{A}^\dagger = \mathbf{A} + d_e^2\frac{\mathbf{j}}{\rho}.$$

Appendix C. Analysis of the smoothing property of the damped-Jacobi iteration for the linearized EMHD system

We specialize the linearized EMHD system in Eq. 23 for a homogeneous uniform plasma in a periodic domain with a constant magnetic field (and therefore no background current) and no electron flow or electron inertia. The corresponding linear system reads:

$$\begin{aligned} \delta\mathbf{A} + \Delta t\,\theta d_i\left(v_A\delta\mathbf{j}\times\mathbf{B_0} - d_i\nu_e\nabla^2\delta\mathbf{j}\right) &= \mathbf{0}, \\ \delta\mathbf{j} - \nabla\times\nabla\times\delta\mathbf{A} &= \mathbf{0}. \end{aligned} \tag{C.1}$$

We have explicitly added the Alfven speed v_A to keep track of dimensionless numbers later. In matrix form, the system can be written as:

$$\begin{bmatrix} \mathbb{I} & \Delta t\,\theta d_i\left[v_A\mathbb{I}\times\mathbf{B}_0 - d_i\nu_e\nabla^2\right] \\ -\nabla\times\nabla\times & \mathbb{I} \end{bmatrix}\begin{bmatrix} \delta\mathbf{A} \\ \delta\mathbf{j} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Calling $\vec{\xi} = (\delta\mathbf{A}, \delta\mathbf{j})^T$, we can write the damped Jacobi iteration as:

$$\vec{\xi}^{\,l+1} = (\mathbb{I} - \sigma D^{-1}A)\vec{\xi}^{\,l}, \tag{C.2}$$

where

$$A = \begin{bmatrix} \mathbb{I} & \Delta t\theta d_i \left[v_A \mathbb{I} \times \mathbf{B}_0 - d_i \nu_e \nabla^2 \right] \\ -\nabla \times \nabla \times & \mathbb{I} \end{bmatrix},$$

D is its diagonal (element- or block-wise), and $\sigma \in (0,1]$ is the damping parameter. For simplicity, we restrict the analysis below to 2D and consider a uniform mesh with $\Delta x = \Delta y = \Delta$ as a sufficiently representative case for the challenges at hand.

In this study, we consider the block diagonal matrix formed by the diagonal coefficients of all vector components at a given cell, with one important approximation: $-\nabla \times \nabla \times \approx \nabla^2$ in the current density equation. As we shall see, this approximation delivers a convergent Jacobi iteration while avoiding the MG difficulties originating in the $\nabla \times \nabla \times$ term without changing the solution to the linear system upon convergence. There results:

$$D = \begin{bmatrix} \mathbb{I} & \Delta t\theta d_i \left[v_A \mathbb{I} \times \mathbf{B}_0 + d_i \nu_e \frac{4}{\Delta^2} \mathbb{I} \right] \\ -\frac{4}{\Delta^2}\mathbb{I} & \mathbb{I} \end{bmatrix} = \begin{bmatrix} \mathbb{I} & U \\ L & \mathbb{I} \end{bmatrix}, \tag{C.3}$$

where:

$$U = \Delta t\theta d_i \left[v_A \mathbb{I} \times \mathbf{B}_0 + d_i \nu_e \frac{4}{\Delta^2} \mathbb{I} \right], \; L = -\frac{4}{\Delta^2}\mathbb{I}.$$

The inverse of the diagonal matrix D is exactly given by:

$$D^{-1} = \begin{bmatrix} \mathbb{I} & -U \\ 0 & \mathbb{I} \end{bmatrix} \begin{bmatrix} \mathbb{I} & 0 \\ 0 & P^{-1} \end{bmatrix} \begin{bmatrix} \mathbb{I} & 0 \\ -L & \mathbb{I} \end{bmatrix} = \begin{bmatrix} \mathbb{I} + UP^{-1}L & -UP^{-1} \\ -P^{-1}L & P^{-1} \end{bmatrix} = \begin{bmatrix} P^{-1} & -\frac{\Delta^2}{4}\left[\mathbb{I} - P^{-1}\right] \\ \frac{4}{\Delta^2}P^{-1} & P^{-1} \end{bmatrix},$$

where:

$$P = \mathbb{I} - LU = \alpha\mathbb{I} + \beta(\mathbb{I} \times \mathbf{B}_0),$$

with:

$$\alpha = 1 + \frac{4\nu_e d_i}{v_A \Delta^2}\beta \; ; \; \beta = \frac{4\Delta t\theta v_A d_i}{\Delta^2}. \tag{C.4}$$

Here, $\beta \gg 1$ is the whistler CFL number (which ensures we are in the stiff hyperbolic limit). The ratio α/β determines the dissipation regime. If we choose ν_e such that dissipation is provided at the grid scale, we have $\nu_e \sim v_A \Delta^2 / d_i$ and therefore $\alpha \sim \beta \gg 1$. Weak and strong hyperresistivity regimes are then defined by $\alpha \ll \beta$ and $\alpha \gg \beta$, respectively. The matrix P has an analytical inverse, given by:

$$P^{-1} = \frac{\alpha\mathbb{I} - \beta(\mathbb{I} \times \mathbf{B}_0) + \frac{\beta^2}{\alpha}(\mathbb{I} \cdot \mathbf{B}_0)\mathbf{B}_0}{\alpha^2 + \beta^2}.$$

To study the convergence properties of the Jacobi iteration, we perform a Von-Neumann analysis with the ansatz:

$$\vec{\xi}^{\,l} = \lambda^l e^{i\mathbf{k}\cdot\mathbf{x}} \vec{\xi}^{\,0},$$

where λ is the Jacobi iteration amplification factor. Stability of the iteration demands $|\lambda| \leq 1$. The smoothing property requires $|\lambda| < 1$ for the high-wavenumber modes in the system. With this ansatz, and for periodic boundary conditions, we can diagonalize all spatial differential operators as:

$$\nabla^2 \;\; \rightarrow \;\; -\left(k_x^2 \mathrm{sinc}^2\left(\frac{k_x\Delta}{2}\right) + k_y^2 \mathrm{sinc}^2\left(\frac{k_y\Delta}{2}\right)\right) = -\kappa_L^2, \tag{C.5}$$

$$\nabla \;\; \rightarrow \;\; i\left(k_x \mathrm{sinc}(k_x\Delta), k_y \mathrm{sinc}(k_y\Delta)\right) = i\boldsymbol{\kappa}_G, \tag{C.6}$$

$$\nabla \times \nabla \times \;\; \rightarrow \;\; -\boldsymbol{\kappa}_G \times \boldsymbol{\kappa}_G \times, \tag{C.7}$$

where $\mathrm{sinc}(\mathrm{x}) = \sin(\mathrm{x})/\mathrm{x}$. The Jacobi (JB) iteration in Eq. C.2 then yields the following equation for the amplification factor λ:

$$(1-\lambda)\begin{bmatrix} \mathbb{I} & 0 \\ 0 & \mathbb{I} \end{bmatrix} = \sigma \begin{bmatrix} \mathbb{I} + K\left[P^{-1} - \mathbb{I}\right] & \frac{\Delta^2}{4}\gamma P^{-1} \\ \frac{4}{\Delta^2} K P^{-1} & \mathbb{I} + \gamma P^{-1} \end{bmatrix}, \tag{C.8}$$

where (using Eq. C.4):

$$\gamma = \frac{d_i \nu_e}{v_A}\beta\left(\kappa_L^2 - \frac{4}{\Delta^2}\right) = (\alpha - 1)\left(\hat{\kappa}_L^2 - 1\right),$$

and the operator K is defined as:

$$K \;=\; \left[\mathbb{I} + \hat{\kappa}_G \times \hat{\kappa}_G \times \mathbb{I}\right],$$

with:

$$\begin{aligned} \hat{\kappa}_L^2 &= \frac{\Delta^2 \kappa_L^2}{4} = \left(\frac{\Delta^2 k_x^2}{4}\mathrm{sinc}^2\left(\frac{k_x\Delta}{2}\right) + \frac{\Delta^2 k_y^2}{4}\mathrm{sinc}^2\left(\frac{k_y\Delta}{2}\right)\right) = \sin^2\left(\frac{k_x\Delta}{2}\right) + \sin^2\left(\frac{k_y\Delta}{2}\right) \in (0,2], \\ \hat{\kappa}_G &= \frac{\Delta \kappa_G}{2} = \frac{1}{2}\left(k_x\Delta\mathrm{sinc}(k_x\Delta), k_y\Delta\mathrm{sinc}(k_y\Delta)\right) = \frac{1}{2}\left(\sin(k_x\Delta), \sin(k_y\Delta)\right). \end{aligned} \tag{C.9}$$

Therefore, Eq. C.8 yields:

$$\begin{bmatrix} (\lambda - 1 + \sigma)\mathbb{I} + \sigma K\left[P^{-1} - \mathbb{I}\right] & \frac{\sigma\Delta^2}{4}\gamma P^{-1} \\ \frac{4\sigma}{\Delta^2} K P^{-1} & (\lambda - 1 + \sigma)\mathbb{I} + \sigma\gamma P^{-1} \end{bmatrix} = 0. \tag{C.10}$$

Equation C.10 is a 2×2 block matrix with commuting blocks. The solution for λ is given by zeroing the determinant of the matrix, which yields a sixth-order characteristic polynomial with six roots. In what follows, we solve for all six roots, and demonstrate that JB is stable and possesses the smoothing property for a suitably chosen damping parameter σ.

The determinant of the 2×2 block system in Eq. C.10 can be written in term of a single block system as (proof via a Schur factorization):

$$\det\left[(\lambda^*)^2 P + \lambda^*\sigma[\gamma\mathbb{I} + K(\mathbb{I} - P)] - \sigma^2\gamma K\right] = 0,$$

where, $\lambda^* = \lambda - 1 + \sigma$. Restricting the analysis to the the stiff hyperbolic regime ($\beta \gg 1$) and to hyperresistivity values large enough to provide dissipation at the grid scale [i.e., $\alpha \geq \beta \gg 1$, which implies $(\mathbb{I} - P) \approx -P$ and $\gamma \approx \alpha\left(\hat{\kappa}_L^2 - 1\right)$], we find:

$$\det\left[(\lambda^*)^2 \frac{P}{\alpha} + \lambda^*\sigma\left[\left(\hat{\kappa}_L^2 - 1\right)\mathbb{I} - K\frac{P}{\alpha}\right] - \sigma^2\left(\hat{\kappa}_L^2 - 1\right)K\right] = 0. \tag{C.11}$$

Here:

$$\frac{P}{\alpha} = \mathbb{I} + \frac{\beta}{\alpha}(\mathbb{I} \times \mathbf{B}_0),$$

and (from Eq. C.4):

$$\frac{\beta}{\alpha} \approx \frac{v_A\Delta^2}{4\nu_e d_i},$$

which is the ratio of the whistler-wave CFL to the hyperresistivity Courant number, and measures the importance of hyperresistivity. Equation C.11 will be the starting point to analyze two dissipation regimes of interest, all of them in the strongly hyperbolic limit ($\beta \gg 1$): the strong hyperresistivity regime ($\alpha \gg \beta$), and the grid-scale hyperresistivity regime ($\alpha \sim \beta$).

Strong hyperresistivity regime: $\alpha \gg \beta \gg 1$. In this case, we have:

$$\left(\frac{P}{\alpha}\right)^{-1} = \alpha P^{-1} \approx \mathbb{I} + \mathscr{O}\left(\frac{\beta}{\alpha}\right).$$

Taking the $\beta/\alpha \to 0$ limit, there results:

$$\det\left[(\lambda^*)^2\mathbb{I} + \lambda^*\sigma\left[\left(\hat{\kappa}_L^2 - 1\right)\mathbb{I} - K\right] - \sigma^2\left(\hat{\kappa}_L^2 - 1\right)K\right] = 0.$$

In vector form, we can write the corresponding linear system as:

$$\left[\lambda^* + \sigma\left(\hat{\kappa}_L^2 - 1\right)\right]\left[\xi\mathbf{A} - K\mathbf{A}\right] = 0, \tag{C.12}$$

with $\xi = \frac{\lambda^*}{\sigma}$. This has the solution (with multiplicity three, and affecting all vector components since it is a prefactor in Eq. C.12):

$$\boxed{\lambda_L = 1 - \sigma\hat{\kappa}_L^2}.$$

The λ_L JB amplification factor is stable and features the smoothing property. In particular, from the definition of $\hat{\kappa}_L^2$ in Eq. C.9, for the Nyquist mode, $k_x = k_x = \pi/\Delta$, we have $\hat{\kappa}_L^2 = 2$ and therefore $\lambda_L = 1 - 2\sigma$, which implies strong damping for a suitably chosen $\sigma \in (0,1)$ (e.g., $\sigma = 0.7$) and therefore a robust smoothing property for error components at the grid scale. Since λ_L has multiplicity three and is affecting all vector components simultaneously, it provides strong smoothing to the whole system.

The remainder of Eq. C.12, $\xi\mathbf{A} = K\mathbf{A}$, corresponds to the $\nabla\times\nabla\times$ operator and is not expected to have the smoothing property since the curl operators are discretized with central differences. However, we show next that the corresponding JB iteration is stable, and therefore will not spoil the smoothing provided by λ_L. Taking components of the remainder of Eq. C.12 and noting that [assuming $B_0 = 1$ without loss of generality and defining $A_\parallel = \mathbf{A}\cdot\mathbf{B}_0$, $A_k = \hat{\kappa}_G\cdot\mathbf{A}$, $A_{\times,\parallel} = \mathbf{B}_0\cdot(\hat{\kappa}_G\times\mathbf{A})$]:

$$\hat{\kappa}_G\cdot K\mathbf{A} = A_k\ ;\ \hat{\kappa}_G\times K\mathbf{A} = K(\hat{\kappa}_G\times\mathbf{A})\ ;\ \mathbf{B}_0\cdot K\mathbf{A} = (1-\hat{\kappa}_G^2)A_\parallel + A_k\hat{\kappa}_{\parallel,G},$$

we find:

- $\mathbf{B}_0\cdot[C.12] \Rightarrow \xi A_\parallel - (1-\hat{\kappa}_G^2)A_\parallel + A_k\hat{\kappa}_{\parallel,G} = 0$,
- $\hat{\kappa}_G\cdot[C.12] \Rightarrow \xi A_k - A_k = 0 \Rightarrow \xi = 1 \Rightarrow \boxed{\lambda = 1}$,
- $\mathbf{B}_0\cdot[\hat{\kappa}_G\times[C.12]] \Rightarrow \xi A_{\times,\parallel} - (1-\hat{\kappa}_G^2)A_{\times,\parallel} = 0 \Rightarrow \boxed{\lambda_G = 1 - \sigma\hat{\kappa}_G^2}$.

The first equation (with $A_k = 0$ from the second equation for $\lambda \neq 1$) also gives $\boxed{\lambda_G = 1 - \sigma\hat{\kappa}_G^2}$. At this point, we have found all six JB roots. As advanced earlier, we conclude that the remaining JB amplification factor roots are stable (since $|\lambda| \leq 1$), but do not feature the smoothing property at the grid scale (since $\hat{\kappa}_G^2 = 0$ for the Nyquist mode and therefore $\lambda_G = 1$).

Grid-scale hyperresistivity regime: $\alpha = \beta \gg 1$. For this case, we have:

$$\det\left[(\lambda^*)^2\mathbb{I} + \lambda^*\sigma\left[\left(\hat{\kappa}_L^2 - 1\right)\alpha P^{-1} - K\right] - \sigma^2\left(\hat{\kappa}_L^2 - 1\right)K\alpha P^{-1}\right] = 0$$

with:

$$\alpha P^{-1} = \frac{\mathbb{I} - (\mathbb{I}\times\mathbf{B}_0) + (\mathbb{I}\cdot\mathbf{B}_0)\mathbf{B}_0}{2}.$$

The corresponding linear system can be expressed in vector form as:

$$(\lambda^*)^2\,\mathbf{A}+\lambda^*\sigma\left[\left(\hat{\kappa}_L^2-1\right)(\alpha P^{-1})\mathbf{A}-K\mathbf{A}\right]-\sigma^2\left(\hat{\kappa}_L^2-1\right)K(\alpha P^{-1})\mathbf{A}=0, \tag{C.13}$$

with:

$$\alpha P^{-1}\mathbf{A}=\frac{\mathbf{A}-\mathbf{A}\times\mathbf{B}_0+A_\parallel\mathbf{B}_0}{2}.$$

Taking components of Eq. C.13, noting that:

- $\hat{\kappa}_G\cdot(\alpha P^{-1})\mathbf{A}=\frac{A_k-A_{\times,\parallel}+A_\parallel\hat{\kappa}_{\parallel,G}}{2}$.
- $\mathbf{B}_0\cdot\left(\hat{\kappa}_G\times(\alpha P^{-1})\mathbf{A}\right)=\frac{A_{\times,\parallel}-A_\parallel\hat{\kappa}_{\parallel,G}+A_k}{2}$.
- $\mathbf{B}_0\cdot(\alpha P^{-1})\mathbf{A}=A_\parallel$.

we find:

- $\mathbf{B}_0\cdot[C.13]\Rightarrow$

$$\left(\left(\frac{\lambda^*}{\sigma}\right)^2+\frac{\lambda^*}{\sigma}\left(\hat{\kappa}_G^2+\hat{\kappa}_L^2-2\right)-\left(\hat{\kappa}_L^2-1\right)\left(1-\hat{\kappa}_G^2\right)\right)A_\parallel-\frac{\lambda^*}{\sigma}A_k\hat{\kappa}_{\parallel,G}-\left(\hat{\kappa}_L^2-1\right)\hat{\kappa}_{\parallel,G}\frac{A_k-A_{\times,\parallel}+A_\parallel\hat{\kappa}_{\parallel,G}}{2}=0.$$

- $\hat{\kappa}_G\cdot[C.13]\Rightarrow\left((\lambda^*)^2-\lambda^*\sigma\right)A_k+\left[\lambda^*\sigma-\sigma^2\right]\left(\hat{\kappa}_L^2-1\right)\frac{A_k-A_{\times,\parallel}+A_\parallel\hat{\kappa}_{\parallel,G}}{2}=0$, or:

$$(\lambda^*-\sigma)\left[\frac{\lambda^*}{\sigma}A_k+\left(\hat{\kappa}_L^2-1\right)\frac{A_k-A_{\times,\parallel}+A_\parallel\hat{\kappa}_{\parallel,G}}{2}\right]=0\Rightarrow\boxed{\lambda=1}\,;\;\frac{\lambda^*}{\sigma}A_k+\left(\hat{\kappa}_L^2-1\right)\frac{A_k-A_{\times,\parallel}+A_\parallel\hat{\kappa}_{\parallel,G}}{2}=0.$$

- $\mathbf{B}_0\cdot[\hat{\kappa}_G\times[C.13]]\Rightarrow$

$$(\lambda^*)^2A_{\times,\parallel}+\lambda^*\sigma\left(\hat{\kappa}_L^2-1\right)\frac{A_{\times,\parallel}-A_\parallel\hat{\kappa}_{\parallel,G}+A_k}{2}-\lambda^*\sigma(1-\hat{\kappa}_G^2)A_{\times,\parallel}-\sigma^2\left(\hat{\kappa}_L^2-1\right)(1-\hat{\kappa}_G^2)\frac{A_{\times,\parallel}-A_\parallel\hat{\kappa}_{\parallel,G}+A_k}{2}=0,$$

or:

$$\left[\lambda^*-\sigma(1-\hat{\kappa}_G^2)\right]\left[\lambda^*A_{\times,\parallel}+\sigma\left(\hat{\kappa}_L^2-1\right)\frac{A_{\times,\parallel}-A_\parallel\hat{\kappa}_{\parallel,G}+A_k}{2}\right]=0,$$

and therefore:

$$\boxed{\lambda_G=1-\sigma\hat{\kappa}_G^2}\,;\;\frac{\lambda^*}{\sigma}A_{\times,\parallel}+\left(\hat{\kappa}_L^2-1\right)\frac{A_{\times,\parallel}-A_\parallel\hat{\kappa}_{\parallel,G}+A_k}{2}=0.$$

We show next that all the vector components computed above feature at least one JB amplification factor with the smoothing property. Introducing the second equation into the first one, we find:

$$A_\parallel\left[\left(\frac{\lambda^*}{\sigma}\right)^2+\frac{\lambda^*}{\sigma}\left(\hat{\kappa}_G^2+\hat{\kappa}_L^2-2\right)+\left(\hat{\kappa}_L^2-1\right)\left(\hat{\kappa}_G^2-1\right)\right]=0.$$

For finite $A_\parallel$, we get:

$$\left(\frac{\lambda^*}{\sigma}\right)^2+\frac{\lambda^*}{\sigma}\left(\hat{\kappa}_G^2+\hat{\kappa}_L^2-2\right)+\left(\hat{\kappa}_L^2-1\right)\left(\hat{\kappa}_G^2-1\right)=0,$$

the solutions of which are:

$$\boxed{\lambda_L=1-\sigma\hat{\kappa}_L^2\,;\,\lambda_G=1-\sigma\hat{\kappa}_G^2},$$

one of which has the smoothing property (λ_L).

Else, $A_{\parallel} = 0$, and from the last two equations we find:

$$\begin{aligned} \lambda^* A_{\times,\parallel} &+ \frac{\sigma\left(\hat{\kappa}_L^2 - 1\right)}{2}\left(A_{\times,\parallel} + A_k\right) = 0, \\ \lambda^* A_k &+ \frac{\sigma\left(\hat{\kappa}_L^2 - 1\right)}{2}\left(A_k - A_{\times,\parallel}\right) = 0. \end{aligned}$$

Solving, we find two complex conjugate roots:

$$\boxed{\lambda_L^{\pm} = 1 - \sigma\left(1 + \frac{\hat{\kappa}_L^2 - 1}{2}\right) \pm i\sigma\frac{|\hat{\kappa}_L^2 - 1|}{2}}.$$

Both these roots have the smoothing property, as we show next. Recall that $|\hat{\kappa}_L^2 - 1| \in [0,1]$, that for stability of the Jacobi iteration we need $|\lambda| \leq 1$, and that for the smoothing property we need the strict inequality for the Nyquist mode. Computing $|\lambda_L^{\pm}|$, we find:

$$|\lambda_L^{\pm}|^2 = (1-\sigma)^2 - (1-\sigma)\sigma(\hat{\kappa}_L^2 - 1) + \sigma^2\frac{(\hat{\kappa}_L^2 - 1)^2}{2}, \tag{C.14}$$

and therefore it follows that:

$$|\lambda_L^{\pm}| < 1 \Rightarrow \sigma < \frac{2(1+\hat{\kappa}_L^2)}{1+\hat{\kappa}_L^4}.$$

This condition is always satisfied, as the right hand side is always greater than unity for $\hat{\kappa}_L^2 \in [0,2]$, and $\sigma \in (0,1]$. Therefore, these roots are stable and strongly damping for the Nyquist mode $\hat{\kappa}_L^2 = 2$. This can be shown by introducing $\hat{\kappa}_L^2$ into Eq. C.14, to find:

$$|\lambda_L^{\pm}|^2 = 1 - 3\sigma + \frac{5}{2}\sigma^2,$$

which is always less than unity for $\sigma \in (0,1]$, and results in very strong damping of the Nyquist mode for a suitably chosen damping parameter: $|\lambda_L^{\pm}|^2 = 0.1$ for $\sigma = 0.6$.

In conclusion, we have again solved for all six roots for the JB amplification factor, and are all stable. While non-smoothing amplification factors (λ_G and $\lambda = 1$) are present for all vector components, they are always accompanied by a robustly smoothing one: λ_L for $A_{\parallel}$ and $\lambda_L^{\pm}$ for $A_{\times,\parallel}$ and A_k. These ensure the smoothing property of the Jacobi iteration for the whole system.